Prime Gaps and Dyadic Tails: Complete Reasoning Record

Erdős Problem #251

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Abstract

Let $p_0 = 2, p_1 = 3, \ldots$ enumerate the primes and let $g_n = p_{n+1} - p_n$. Both dyadic series converge and

$$\Pi = \sum_{n \geq 0} \frac{p_n}{2^{n+1}} = 2 + \sum_{n \geq 0} \frac{g_n}{2^{n+1}} = 2 + S.$$

The identity follows from summation by parts and the elementary polynomial bound $p_n \leq 1250(n+1)^4$ proved in Appendix A. The substantive limitation comes from rationalising perturbations: bounded residue-preserving changes to the actual gaps can make their dyadic value rational. The record recovers this construction, the exact tail criteria, and the finite denominator exclusions; the identity provides their common normalisation.

Write $T_N = \sum_{j \geq 1} g_{N+j} 2^{-j}$ for the scaled tails, so that $T_{N+1} = 2T_N - g_{N+1}$. Irrationality of Π is equivalent to nonintegrality of every positive tail shift $T_{N+h} - T_N$, and equivalently to the free-pair condition that for every modulus t and every cutoff some pair of indices congruent modulo t beyond the cutoff has nonintegral tail difference. Two adjacent shifts in $(-1, 1)$ with unequal corresponding gaps cannot both be integral, and an explicit remainder bound certifies one such pair at $h = 1$, $N = 2$ by exact integer arithmetic. Every rational equal to Π, and hence every rational equal to S, has denominator at least $2^{589} > 10^{177}$ by a certificate the Lean kernel decides, and at least $2^{39997} > 10^{12040}$ by a certified continued-fraction prefix.

Four proved statements constrain any argument. Raising every gap by at most M beyond a prescribed prefix, preserving each residue modulo M, already makes the sum rational. That rationalising perturbation inherits the fixed-block polynomial nonconcentration property which Schlage-Puchta proved for the actual gaps, so nonconcentration together with prefix, size, residue and cumulative-growth data still permits a rational value. Under the two window conditions the gap mismatch is forced to be ± 2, and by the same nonconcentration lemma that event has density zero, so the missing input is a cofinality statement on a sparse set. Finally, two even values differing by 2, each recurring infinitely often inside every index residue class, do not force irrationality: an explicit positive even word of prime-number-theorem cumulative growth has both values recurring and an integral tail at every index. Problem #251 remains open, and the exact unproved input is cofinal nonintegrality of the actual prime-gap tail shifts.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

The exact identity, the finite exclusions, and the open boundary

Exact identity. The prime-index dyadic series equals two plus the consecutive-prime-gap dyadic series, unconditionally, with summability and the identity checked by the Lean kernel over an elementary polynomial prime bound. **Denominator floors.** Every rational equal to either series has denominator at least $2^{589} > 10^{177}$ by a kernel-decided certificate, and at least $2^{39997} > 10^{12040}$ by a certified continued-fraction prefix. **Exact criteria.** Irrationality is equivalent to cofinal nonintegrality of every fixed positive tail shift, to the free-pair condition with the offset free, and to nonintegrality along the lcm diagonal. **Proved obstructions.** Bounded residue-preserving perturbations make the sum rational; that perturbation inherits fixed-block polynomial nonconcentration; the two-window event forces a ± 2 mismatch and therefore has density zero; and recurring gap values differing by 2 inside every residue class are compatible with an integral tail at every index. **Open boundary.** No result here proves cofinal nonintegrality for the actual prime gaps or irrationality of either series, and Erdős #251 remains open.

1 Introduction

Let p_n denote the nth prime. Erdős Problem #251 asks whether

$$\Pi = \sum_{n \geq 1} \frac{p_n}{2^n}$$

is irrational [1, p. 94][2, p. 62][4, p. 103]. Bloom's catalogue records the problem as open [16]. Numerically $\Pi = 2/2 + 3/4 + 5/8 + 7/16 + 11/32 + \cdots = 3.674643966\ldots$.

Throughout the formal development the primes are indexed from zero, so that $p_0^{(0)} = 2$, $p_1^{(0)} = 3$, $p_2^{(0)} = 5$, and the gaps are $g_i = p_{i+1}^{(0)} - p_i^{(0)}$. In that indexing

$$\Pi = \sum_{i \geq 0} \frac{p_i^{(0)}}{2^{i+1}},$$

which is the convention every statement below uses; we drop the superscript from here on. The first values are

i	0	1	2	3	4	5	6	7	8	9
p_i	2	3	5	7	11	13	17	19	23	29
g_i	1	2	2	4	2	4	2	4	6	2

so $g_0 = 1$ and every later gap is even, the primes after 2 being odd. A second normalisation with denominator 2^i also occurs, and at every finite horizon it is exactly twice this one,

$$\sum_{i=0}^{n-1} \frac{p_i}{2^i} = 2 \sum_{i=0}^{n-1} \frac{p_i}{2^{i+1}},$$

the factor-of-two normalisation. A factor of two does not change rationality, so no result below depends on the choice; the identity is stated so that the indexing cannot drift silently.

Notation. $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\mathbb{N}_{>0} = \{1, 2, \dots\}$. We write φ for Euler's totient function, $\operatorname{den} q$ for the denominator of a rational number q written in lowest terms, $\operatorname{dist}(x, \mathbb{Z})$ for the distance from a real number x to the nearest integer, and $\operatorname{Irr}(x)$ for the assertion that x is irrational. A rational number is called *integral* when it is the image of an integer. A sequence (a_n) is *eventually periodic with period* $h \ge 1$ when $a_{n+h} = a_n$ for every sufficiently large n. A set $A \subseteq \mathbb{N}$ has *density zero* when $|A \cap [1, x]| = o(x)$, and a property holds for *almost all* indices when its exceptional set has density zero. A sum over an empty range of indices is zero.

Relation to prior work. The passage from the primes to their gaps is already public. Tao recorded on the problem page that summation by parts makes the question equivalent to irrationality of $\sum_n (p_{n+1} - p_n) 2^{-n}$, and named the shape of the missing input as a sufficiently quantitative and uniform prime-tuples hypothesis giving statistical control of the binary expansion of about $\log\log n$ consecutive gaps [17, comment of 7 October 2025]. Theorem 2.3 below is that reduction in exact form, with the endpoint retained, with convergence supplied by an elementary bound, and with the statement checked by the Lean kernel. The elementary bound replaces the prime number theorem. No priority is claimed for the reduction.

Land has since posted a conditional proof of irrationality assuming Kuperberg's uniform Hardy-Littlewood prime-tuples conjecture, together with a Lean formalisation of that conditional argument [9, Conjecture 1 and §1]. His route constructs weighted prime-gap tails converging to 6 from above, imposes a consecutive quadratic prime pattern by a mixed prime-counting moment, and averages over later prime indices to control the unprescribed tail. Nothing in this note is conditional on a prime-tuples hypothesis, and nothing in this note proves the unconditional statement his argument would need.

Erdős stated on p. 93 of the 1958 article that $\sum_n p_n^k / n!$ is irrational for every $k \ge 1$, and wrote there that the proof for $k > 1$ is complicated enough that only the case $k = 1$ is printed [1, pp. 93–95]. A proof of the full family appears in Schlage-Puchta's Theorem 3, which gives the stronger statement that $1, S_0, S_1, S_2, \dots$ are linearly independent over $\mathbb{Q}$, where $S_k = \sum_{n \ge 1} p_n^k / n!$ [14, Theorem 3]. The catalogue entry still attributes the whole family to the 1958 article [16].

On page 103 of his 1988 problem paper Erdős separately stated the fixed-denominator problem: he could not prove that $\sum_n p_n^k / 2^n$ is irrational for every $k \ge 1$, and wrote that the case $k = 1$ was probably already very difficult. He also stated the variable-denominator expectation that $\sum_{n \ge 1} p_n / (g_1 \cdots g_n)$ is irrational whenever $g_n \ge 2$ and $g_n = o(p_n)$ [4, p. 103]. That expectation is false: Kovač posted a construction of such a sequence (g_n) for which the sum is exactly 1, on 15 April 2026 [18, Theorem 1 and proof, pp. 1–2]. The catalogue record still repeats the refuted expectation without mentioning the counterexample, as checked on 6 September 2026 [16]; its warning that relevant literature may be missing applies directly to this omission.

Section 3 of the 1958 article proves a classification for the variable-denominator family, and the hypotheses matter. Let $1 < q_1 \le q_2 \le \cdots$ be integers satisfying, for some $k > 0$, the lower growth condition $q_n > o(n / \log^k n)$. Then $\sum_{n \ge 1} p_n / (q_1 \cdots q_n)$ is rational if and only if $q_n = q\, p_n + 1$ for one fixed integer $q \ge 1$ and all large n [1, pp. 96–97]. The denominators are nondecreasing, and the printed condition bounds them from below. Strict increase is not assumed, and no upper bound is imposed. The endpoint $q = 1$ is attained: if $G_n = \prod_{j=1}^{n}(p_j + 1)$ then $p_n/G_n = 1/G_{n-1} - 1/G_n$, so the corresponding series telescopes to 1. Kovač's sequence evades the lower growth condition, and neither statement addresses the fixed-denominator dyadic series studied here.

There is also a genuinely adjacent proved dyadic theorem. If $P(m)$ denotes the largest prime factor of m, Erdős and Pomerance proved that

$$\sum_{m\geq 2} \frac{\mathbf{1}_{\{P(m)>P(m+1)\}}}{2^m}$$

is irrational [3, §7, p. 320]. Erdős and Graham record the complementary indicator on p. 62: equality of the two largest prime factors is impossible for consecutive integers, so its series is $1/2$ minus the displayed one and is irrational as well. That is a theorem about a bounded prime-factor comparison digit sequence, and the numerators in Π are unbounded.

Two further results of Schlage-Puchta enter below. His Theorem 2 classifies rationality for numbers whose base-b digit string concatenates the representations of a slowly growing integer sequence, and is a different object from the tail criterion here; it was already pointed at this problem in the catalogue thread on 15 April 2026. His Lemma 4 is the input this note actually uses. It states that for every polynomial $F \in \mathbb{Z}[x_0, \dots, x_k]$ which does not vanish identically, $F(g_n, \dots, g_{n+k}) \neq 0$ for almost all n, where g_n are the actual consecutive prime gaps [14, Lemma 4, pp. 5–6]. Its proof runs through Selberg's sieve. Sections 6 draws two consequences from it.

Problem #251 also appears as the unproved declaration `erdos_251` in the *Formal Conjectures* repository [19]. Its zero-based Lean sum starts with the zeroth prime over 2^0, so it is twice the displayed normalisation and has equivalent irrationality status; its proof is `sorry`.

The strategy. The digits p_n grow, so the series is not a digit expansion in any bounded alphabet, and the standard rationality criteria for such expansions do not apply directly. The classical elementary criteria for series of this kind instead control irrationality through the growth of the denominators. Erdős and Straus named a sum $\sum_k 1/a_k$ over a strictly increasing sequence of positive integers an *Ahmes series* [5]; for such a series the condition $a_k^{1/2^k} \to \infty$ is sufficient for irrationality, and it is sharp, since shifted Sylvester sequences grow like C^{2^k} for arbitrarily large C and have rational reciprocal sum. Both statements and their attribution are recorded in the introduction of Kovač and Tao [8, §1]. Splitting each term $p_i/2^{i+1}$ into p_i copies of $2^{-(i+1)}$ writes Π as a sum of unit fractions with repetitions, and the denominators occurring in it are exactly the powers of two. Even ignoring the repetitions the growth hypothesis fails by every available margin, since $(2^n)^{1/2^n} \to 1$, and every arithmetic constraint has to come from the numerators instead.

What replaces growth control is denominator control on the sequence of rescaled tails. Rationality of the sum is equivalent to an eventual integrality condition on differences of those tails, and that condition uses nothing about the numerators beyond the fact that they are integers. The condition does not by itself force the numerators to repeat: Proposition 6.9, applied to $K_n = n$, produces the integer sequence $\kappa_n = n - 1$, which is unbounded and hence not eventually periodic, and whose dyadic sum is zero.

Outline. Section 2 proves the identity and the irrationality equivalence. Section 3 develops the tail recurrence and states the three exact criteria: pointwise shift escape, the free-pair form with a free offset, and the lcm diagonal. Section 4 gives the local obstruction, its signed normal form, the explicit remainder bound and one certified actual pair. Section 5 records the two denominator floors. Section 6 proves the four obstructions. Section 7 reports the two finite measurements. Section 8 states the remaining obligation. All declarations linked below are checked by the Lean 4 kernel against the pinned Mathlib revision at the formal-source commit ee650b32b8b2, and the pinned sources contain no proof placeholders and no project-defined axioms. The cited system paper identifies Lean 4 [10, abstract and §1, pp. 625–626], while the Mathlib paper documents the library's historical Lean 3-era architecture [11, abstract and §1.1, p. 367]; it is not authority for the current pinned revision.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no `sorry`, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Companion system context. The claim and trust boundary, cold-clone route to proof authority, and public contribution protocol are described in sibling papers. Those descriptions do not change the mathematical status of this note.

Statement	Status	Treatment here
Irrationality of Π	Open	Not proved.
Finite prime-gap identity	Proved here	Theorem 2.2, with the endpoint retained.
Infinite prime-gap identity	Lean-checked unconditionally	Theorem 2.3.
Prime-series/gap-series irrationality equivalence	Lean-checked unconditionally	Corollary 2.4.
Elementary polynomial prime bound $p_n \le 1250(n+1)^4$	Lean-checked; proof reprinted here	Appendix A.
Actual real prime-gap tail and its recurrence	Lean-checked unconditionally	Section 3.
Rationality/integral-shift classification	Lean-checked; an equivalence	Theorem 3.3.
Free-pair irrationality criterion	Lean-checked; an equivalence	Theorem 3.4.

Lcm-diagonal irrationality criterion	Lean-checked; an equivalence	Theorem 3.5.
Adjacent small-shift obstruction, rational and real	Lean-checked	Theorem 4.1, Corollary 4.2.
Signed two-window normal form	Lean-checked; an equivalence	Theorem 4.4.
Explicit remainder bound and the certified pair at $h = 1, N = 2$	Ordinary proof; exact integer replay in Appendix B	Propositions 4.5 and 4.6.
Denominator floor $b \geq 2^{589}$	Lean-checked by `decide +kernel`; ordinary proof reprinted here	Theorem 5.1.
Certified continued-fraction exclusion $q \geq 2^{39997}$	Exact computation over a Lean-checked tail bound	Theorem 5.2; no Lean declaration.
Bounded-perturbation obstruction	Lean-checked	Theorem 6.1.
Sparse rationalisation with cumulative congruences	Ordinary theorem; elementary Lean core only	Theorem 6.2; upper Banach density zero, arbitrarily slow eventual coefficient bound, and eventual divisibility of coefficients and cumulative corrections. The explicit schedule is proved in the paper.
Nonconcentration survives the rationalising perturbation	Ordinary proof; external lemma cited	Theorem 6.4 and Corollary 6.5.
The two-window event has density zero	Ordinary proof; external lemma cited	Theorem 6.6.
Recurring gap values differing by 2 do not suffice	Ordinary proof	Theorem 6.7.
Quadratic polynomial-shift countermodel	Lean-checked	Proposition 6.8.
Rationality alone forces periodic integer coefficients	False	Proposition 6.9.
Affine and fixed-lattice cofinal escape tests	Lean-checked; both equivalences	Theorem 6.10.
Actual prime gaps are unbounded and not eventually periodic	Lean-checked	Proposition 4.3.

Adjacent-mismatch event density below 1.2×10^8	Finite measurement, floating-point tails	Section 7.
Free-pair witnesses for every $t \le 20$ below 2×10^7	Finite measurement	Section 7.
Cofinal adjacent small mismatch	Sufficient condition; not proved	Problem 8.2.

How to read the middle column. Proved here marks a statement proved in the text. *Lean-checked* marks a statement accepted by the kernel at the pinned commit, with the declaration reachable through the linked phrase. *Exact computation* marks a finite computation with a recorded receipt and no Lean declaration. *Ordinary proof* marks a derivation written out here with no formal check. *Finite measurement* marks a computation over one bounded range which supplies no cofinal statement.

Keywords. irrationality; prime gaps; dyadic series; summation by parts; Lean 4. **MSC 2020.** 11J72 (primary); 11N05, 68V20 (secondary).

2 Summation by parts, with the endpoint retained

Summation by parts trades a sequence for its consecutive differences. The form recorded here is exact: it carries no error term, it assumes nothing about the sequence, and it retains the endpoint term. No endpoint is absorbed into an estimate. For a sequence P of rational numbers and $n \ge 0$ put

$$D(P,n) = \sum_{i=0}^{n-1} \frac{P(i)}{2^{i+1}}, \qquad \Delta(P,n) = \sum_{i=0}^{n-1} \frac{P(i+1) - P(i)}{2^{i+1}},$$

both empty, hence zero, at $n = 0$. We call these the dyadic partial sum and the dyadic difference sum of P.

Proposition 2.1 (finite summation by parts)**.** *For every rational sequence P and every $n \ge 0$,*

$$D(P,n+1) = P(0) + \Delta(P,n) - \frac{P(n)}{2^{n+1}}.$$

Proof. A routine induction on n. At $n = 0$ both sides equal $P(0)/2$. For the step, adding $P(n+1)/2^{n+2}$ to the left and $(P(n+1) - P(n))/2^{n+1}$ to the difference sum changes the endpoint term from $P(n)/2^{n+1}$ to $P(n+1)/2^{n+2}$, and the two adjustments agree. □

Formalised as the summation-by-parts identity. Nothing is assumed about P: no positivity, no monotonicity, and no convergence.

Specialising to $P(i) = p_i$, whose first value is $p_0 = 2$, and writing $g_i = p_{i+1} - p_i$ for the zero-based gaps, gives the reformulation.

Theorem 2.2 (prime-gap reformulation)**.** *Let $p_0 = 2, p_1 = 3, \ldots$ be the primes in increasing order and $g_i = p_{i+1} - p_i$. For every $n \ge 0$,*

$$\sum_{i=0}^{n} \frac{p_i}{2^{i+1}} = 2 + \sum_{i=0}^{n-1} \frac{g_i}{2^{i+1}} - \frac{p_n}{2^{n+1}}.$$

Formalised as the prime-gap summation by parts, using the gap partial sum and the gap cast identity; the latter records that the natural-number difference $p_{n+1} - p_n$ agrees with the difference taken in $\mathbb{Q}$, which needs $p_n \le p_{n+1}$, the monotonicity step. The leading 2 is the first prime, not a normalising constant. At $n = 2$, for instance, the left side is $2/2 + 3/4 + 5/8 = 19/8$ and the right side is $2 + (1/2 + 2/4) - 5/8 = 19/8$.

2.1 The infinite identity and the irrationality equivalence

Write

$$u_n = \frac{p_n}{2^{n+1}}, \qquad v_n = \frac{g_n}{2^{n+1}}$$

for the terms of the prime series and of the gap series, so that $\sum_{n\ge 0} u_n = \Pi$. The termwise identity $v_n = 2u_{n+1} - u_n$ is the dyadic discrete derivative. It expresses each gap term as an integer combination of two consecutive prime terms, so once (u_n) is summable the gap series can be summed by rearranging two copies of the prime series.

Theorem 2.3 (infinite prime-gap identity)**.** *Both series converge and*

$$\sum_{n\ge 0} \frac{p_n}{2^{n+1}} = 2 + \sum_{n\ge 0} \frac{g_n}{2^{n+1}}.$$

Proof. The bound $p_n \le 1250(n+1)^4$ of Appendix A makes (u_n) summable, and $0 < g_n \le p_{n+1}$ then makes (v_n) summable. The shifted sequence (u_{n+1}) is summable, so summing $v_n = 2u_{n+1} - u_n$ and using $u_0 = 1$ gives

$$\sum_{n\ge 0} v_n = 2\left(\sum_{n\ge 0} u_n - u_0\right) - \sum_{n\ge 0} u_n = \sum_{n\ge 0} u_n - 2. \qquad \square$$

The polynomial bound is checked here, summability of the prime series here, the summability transfer is the gap-series summability theorem, and the displayed identity is the unconditional prime-gap identity. The prime number theorem is not a dependency at any point.

Corollary 2.4 (exact irrationality reformulation)**.** Π *is irrational if and only if* $S = \sum_{n\ge 0} g_n 2^{-(n+1)}$ *is irrational. The corresponding zero-based series with denominator* 2^n *equals* $4 + 2S$ *and has the same irrationality status.*

These are the normalised irrationality equivalence, the displayed-series identity, and the displayed-series irrationality equivalence. Concretely, $\Pi = 3.674643966\ldots$ is irrational if and only if $S = 1.674643966\ldots$ is, and neither is known.

3 The tail recurrence and the exact criteria

The series is not attacked directly. Suppose $\sum_{i\ge 0} a_i 2^{-(i+1)}$ converges with every a_i an integer, and rescale its tails by putting

$$T_N = 2^{N+1} \sum_{i>N} \frac{a_i}{2^{i+1}} = \sum_{j\ge 1} \frac{a_{N+j}}{2^j}.$$

Two facts follow immediately. First $T_{N+1} = 2T_N - a_{N+1}$, so moving one level along doubles the rescaled tail and subtracts a single coefficient. Second $T_0 = 2\sum_{i\geq 0} a_i 2^{-(i+1)} - a_0$, so the sum is rational exactly when T_0 is. This section studies that recurrence, assuming nothing about the coefficients beyond the fact that they are integers, and resumes the prime-gap instance at the end.

Definition 3.1. Let $g : \mathbb{N} \to \mathbb{Z}$ and $T : \mathbb{N} \to \mathbb{Q}$ or $T : \mathbb{N} \to \mathbb{R}$. Say T satisfies the *dyadic tail recurrence* with *digits* g when $T_{N+1} = 2T_N - g_{N+1}$ for every N. Write $\sigma_h(N) = T_{N+h} - T_N$ for the *shift* of length h at N, and call a rational number *integral* when it is the image of an integer.

These are the tail recurrence, the shift, and integrality. One small orbit is worth having in view. Take every digit $g_N = 0$ and $T_0 = 1/12$. Then the orbit is $\frac{1}{12}, \frac{1}{6}, \frac{1}{3}, \frac{2}{3}, \frac{4}{3}, \ldots$ and $\sigma_h(N) = (2^h - 1)2^N/12$. The shift of length 2 is not integral at $N = 0$ and is integral at $N = 2$; the shift of length 1 is never integral. Nonintegrality at one prescribed shift length therefore does not imply irrationality.

Iterating the recurrence h times multiplies T_N by 2^h and accumulates an explicit integer. Define $B_{0,N} = 0$ and $B_{h+1,N} = 2B_{h,N} + g_{N+h+1}$, so that $B_{h,N} = g_{N+1}2^{h-1} + \cdots + g_{N+h}$: the tail block.

Theorem 3.2 (block identity)**.** *For every N and h,*

$$T_{N+h} = 2^h T_N - B_{h,N}, \qquad \textit{hence} \qquad \sigma_h(N) = (2^h - 1)\, T_N - B_{h,N}.$$

Proof. A routine induction on h; the step is one application of the recurrence together with the recursion defining B. The second identity is the first minus T_N. □

Formalised as the iterated block identity and the scaled shift identity. The shift also obeys the recurrence in its own right,

$$\sigma_h(N+1) = 2\sigma_h(N) - (g_{N+h+1} - g_{N+1}), \tag{1}$$

the shift step identity. Since $B_{h,N}$ is an integer, Theorem 3.2 converts a question about the shift into a question about $(2^h - 1)T_N$: the shift $\sigma_h(N)$ is integral if and only if $(2^h - 1)T_N$ is, the integral-shift criterion. The criterion holds in both directions, so once T_N is fixed no choice of the digits after index N can change whether $\sigma_h(N)$ is an integer.

For a rational orbit the denominator settles the matter exactly:

$$\operatorname{den}(T_{N+1}) = \frac{\operatorname{den}(T_N)}{\gcd(2, \operatorname{den}(T_N))}, \qquad \sigma_h(N) \in \mathbb{Z} \iff \operatorname{den}(T_N) \mid 2^h - 1. \tag{3.1}$$

Each even denominator loses exactly one factor of 2 and an odd denominator is unchanged, so after finitely many steps the denominator is an odd integer d; Euler's congruence then gives $d \mid 2^{\varphi(d)} - 1$, and the multiplicative order of 2 modulo d is the least such exponent. These are the denominator recurrence, its odd case and even case, the denominator classification, the totient shift, and the propagation of an integral shift to every later index. The hypothesis that d is odd cannot be dropped: if $T_N = 1/2$ then $2^h - 1$ is odd for every $h \geq 1$, so no shift at N is integral.

Theorem 3.3 (exact rationality classification). *Let $T : \mathbb{N} \to \mathbb{R}$ satisfy $T_{N+1} = 2T_N - g_{N+1}$ with integer digits g. The following are equivalent:*

(i) *T_0 is rational;*

(ii) *$\sigma_h(N)$ is an integer for some $h \geq 1$ and some N;*

(iii) *for some fixed $h \geq 1$, $\sigma_h(N)$ is an integer at every sufficiently large N.*

Consequently T_0 is irrational if and only if every positive-length shift is nonintegral at every index, equivalently if and only if for every $h \geq 1$ and every cutoff some later N has $\sigma_h(N) \notin \mathbb{Z}$.

Proof. For (i)⇒(iii), the block identity identifies the whole orbit with the cast of the rational recurrence starting at T_0; apply (3.1) at an index where the denominator has become odd, and propagate forward. The implication (iii)⇒(ii) is immediate. For (ii)⇒(i), the block identity gives $\sigma_h(N) = (2^h - 1)T_N - B_{h,N}$ with $B_{h,N}$ and $\sigma_h(N)$ integers and $2^h - 1 \neq 0$, so T_N is rational, and $T_N = 2^N T_0 - B_{N,0}$ then makes T_0 rational. Negating the three conditions gives the two irrationality formulations. □

The exact real classifiers are one integral positive shift, eventual integrality, pointwise nonintegrality, and cofinal nonintegrality.

The actual prime-gap orbit. Put

$$T_N = \sum_{j \geq 1} \frac{g_{N+j}}{2^j},$$

which converges by Theorem 2.3. The scaled tail equals its shifted-gap series, the shifted-gap series identity, and satisfies $T_{N+1} = 2T_N - g_{N+1}$ with no rationality hypothesis, the real tail recurrence. Since $T_0 = 2S - 1 = 2.349287932\ldots$, the numbers Π, S and T_0 have the same rationality status, and Theorem 3.3 specialises to the actual series: irrationality of Π is equivalent to cofinal nonintegrality of the positive shifts of T, the escape equivalence. Nothing about the bridge from the series to its actual tail is left open; the open content is the escape hypothesis itself.

3.1 The free-pair criterion and the lcm diagonal

The escape condition quantifies over a fixed shift length. Freeing the offset gives an equivalent condition with a weaker-looking producer.

Theorem 3.4 (free-pair criterion). *S is irrational if and only if for every $t \geq 1$ and every N_0 there are $N, M \geq N_0$ with $M \equiv N \pmod{t}$ and $T_M - T_N \notin \mathbb{Z}$.*

Proof. If S is irrational, take $M = N + t$ for the N supplied by Theorem 3.3 at shift length t. Conversely, suppose S is rational. By (3.1) the orbit reaches an index N_0 beyond which the reduced denominator is a fixed odd d; let t be the multiplicative order of 2 modulo d. For $N, M \geq N_0$ with $M \equiv N$ modulo t, the difference $T_M - T_N$ is then an integer, contradicting the stated property at this t and this cutoff. □

The offset $M - N$ is free, so one nonintegral congruent pair for each modulus and each cutoff suffices in place of a supply at a single fixed offset. The mechanism is exact: beyond a rational state with odd reduced denominator d, the difference $T_M - T_N$ is integral precisely when $M \equiv N$ modulo the order of 2 in $\mathbb{Z}/d\mathbb{Z}$, the free-pair lattice, so every rational orbit has a cutoff and a modulus at which integrality holds exactly on the congruence classes, the lattice corollary. The criterion for the actual series is checked as the free-pair equivalence, over the actual real tail recurrence. No antecedent for this packaging has been located; the nearest located neighbour is Schlage-Puchta's Theorem 2 [14], which treats base-b digit strings formed by concatenating a slowly growing integer sequence under a rationality hypothesis, and is a different object, and the sweep of the dyadic tail-recurrence and Bézivin-style functional-equation literature has not been run.

The criterion rules out two shortcuts. For the actual prime-gap tail, Section D.4 combines the prime number theorem with a first-moment bound to show that, under rationality, some tail value occurs $\gg_{C,d} X/\log X$ times in $(X, 2X]$; here d is the eventual odd reduced denominator and $C > 1$ is fixed. This is specific to the prime tail, rather than a consequence of rationality for arbitrary integer-digit orbits, and equal tails have integral difference, so they do not supply the free-pair producer. Also, under the free-pair lattice the difference is an integer whenever the modulus divides the offset, and an integer cannot lie in $(\frac{1}{2}, 1)$, so the window component of the reduced event below is already a contradiction on its own. Both derivations are ordinary proofs, recorded in the extended record and not formalised.

A second exact reformulation collapses all shift lengths and basepoints onto a single schedule. Let $L_0 = 1$ and $L_j = \operatorname{lcm}(1, \ldots, j)$.

Theorem 3.5 (lcm-diagonal criterion)**.** *Let $g : \mathbb{N} \to \mathbb{Z}$ and $T : \mathbb{N} \to \mathbb{R}$ satisfy $T_{N+1} = 2T_N - g_{N+1}$. Then T_0 is irrational if and only if $T_{2L_j} - T_{L_j} \notin \mathbb{Z}$ for every $j \geq 0$.*

Proof. Theorem 3.3 gives the forward implication. Conversely, if T_0 is rational then some positive shift length h is integral at every basepoint beyond an index N_0. Choose j so large that $N_0 \leq L_j$ and $h \mid L_j$. The cocycle identity $\sigma_{a+b}(N) = \sigma_a(N) + \sigma_b(N + a)$ shows by induction that every positive multiple of h is integral at every such basepoint, so $T_{2L_j} - T_{L_j}$ is an integer. □

The factorial schedule has the same divisibility property; the lcm schedule is pointwise no larger and is the endpoint used here. Theorems 3.4 and 3.5 are exact reformulations on the class of integer-digit orbits. They change the shape of the required producer and they supply no nonintegrality statement for the actual prime gaps.

4 A local certificate, and one actual pair

The classifiers reduce irrationality to a condition of infinite precision imposed on a complete tail. Inside $(-1, 1)$ integrality is equality with zero, so on that range the shift step identity (1) turns simultaneous integrality of two adjacent shifts into a single comparison of digits. What this buys is a certificate attached to one pair of adjacent indices, whose verification already contradicts integrality there. The distance of a shift from the integers never has to be estimated.

Theorem 4.1 (adjacent small-shift obstruction)**.** *Let T satisfy the dyadic tail recurrence with integer digits g, and fix h and N. If*

$$-1 < \sigma_h(N) < 1, \qquad -1 < \sigma_h(N+1) < 1, \qquad g_{N+h+1} \neq g_{N+1},$$

then $\sigma_h(N)$ and $\sigma_h(N+1)$ cannot both be integers. Consequently, if such a pair occurs beyond every threshold, the h-shift is not eventually integral.

Proof. An integer strictly between -1 and 1 is zero. If both shifts were integers, both would vanish, and (1) would give $g_{N+h+1} = g_{N+1}$. The cofinal statement chooses one such adjacent pair after the alleged onset of integrality. □

Corollary 4.2 (real form and the sufficient condition)**.** *The same statement holds for a real orbit, with the same proof. If for every $h \geq 1$ and every cutoff some later N satisfies the three displayed conditions for the actual prime gaps, then Π is irrational.*

The rational finite contradiction is the adjacent small-shift obstruction with its cofinal form; the real form is the real adjacent-pair consumer, and the implication to irrationality of the actual series is the small-mismatch criterion. The third hypothesis alone is available for the actual gaps: it reads $g_4 = 2 \neq 4 = g_3$ at $h = 1$, $N = 2$, and by Proposition 4.3 it holds for arbitrarily large N at every fixed h. What is missing is the pair of inequalities, each of which constrains a complete infinite tail.

Proposition 4.3 (prime gaps do not become periodic)**.** *For every positive h, the actual consecutive-prime-gap sequence is not eventually periodic with period h.*

Proof. For $m \geq 2$ the integers $(m+1)!+2, \ldots, (m+1)!+m+1$ are composite, so the gaps are unbounded. An eventually periodic sequence of natural numbers has finite range after its preperiod and a bounded initial segment, hence is bounded. □

Lean checks the factorial argument as unboundedness of the actual gaps and the conclusion as non-eventual periodicity. Far stronger lower bounds for large gaps are known [6, Theorem 1, p. 66]; unboundedness is all that is needed.

The conjunction in Theorem 4.1 has an exact normal form. For fixed h write $D_N = \sigma_h(N)$ and $\delta_N = g_{N+h+1} - g_{N+1}$, so that $D_{N+1} = 2D_N - \delta_N$.

Theorem 4.4 (signed two-window normal form)**.** *Assume δ_N is even. The conjunction $-1 < D_N < 1$, $-1 < D_{N+1} < 1$, $\delta_N \neq 0$ is equivalent to*

$$(\delta_N = 2 \text{ and } \tfrac{1}{2} < D_N < 1) \quad \text{or} \quad (\delta_N = -2 \text{ and } -1 < D_N < -\tfrac{1}{2}).$$

Proof. The recurrence and the two unit windows give $-3 < \delta_N < 3$. A nonzero even integer in that interval is 2 or -2. Substitution into $D_{N+1} = 2D_N - \delta_N$ gives the stated half-window and, in the reverse direction, recovers the second unit window. □

For the actual gaps and $N \geq 1$ every δ_N is even, so Theorem 4.4 applies throughout the range where the local certificate is sought. The same computation for a real orbit is one line and is used in Theorem 6.6 below.

4.1 An explicit remainder and a certified actual pair

Both window conditions involve complete infinite tails. A dominated truncation reduces each of them to a finite integer comparison.

Proposition 4.5 (explicit remainder). *For integers $h, N \geq 0$ and $L \geq 1$ put*

$$F_{h,N,L} = \sum_{j=1}^{L} \frac{g_{N+h+j} - g_{N+j}}{2^j}, \qquad P(x) = x^4 + 8x^3 + 36x^2 + 104x + 150,$$

$$E_{h,N,L} = \frac{1250}{2^L}\big(P(N+h+L+2) + P(N+L+2)\big).$$

Then $|\sigma_h(N) - F_{h,N,L}| \leq E_{h,N,L}$. In particular $|F_{h,N,L}| + E_{h,N,L} < 1$ certifies $|\sigma_h(N)| < 1$, and $\operatorname{dist}(F_{h,N,L}, \mathbb{Z}) > E_{h,N,L}$ *certifies $\sigma_h(N) \notin \mathbb{Z}$.*

Proof. The checked bound $p_n \leq 1250(n+1)^4$ gives $0 \leq g_m \leq p_{m+1} \leq 1250(m+2)^4$, so the omitted absolute tail is at most

$$1250 \sum_{j>L} \frac{(N+h+j+2)^4 + (N+j+2)^4}{2^j}.$$

The polynomial identity $2P(x) = (x+1)^4 + P(x+1)$ telescopes to $\sum_{k=1}^{J}(m+k)^4 2^{-k} = P(m) - P(m+J)2^{-J}$, whose last term tends to zero. Apply it with $m = N+h+L+2$ and $m = N+L+2$ after writing $j = L+k$. The two tests follow from the triangle inequality and the definition of distance to $\mathbb{Z}$. ☐

Proposition 4.6 (a certified adjacent pair). *For the actual prime gaps, $h = 1$ and $N = 2$ satisfy the three hypotheses of Theorem 4.1: both $\sigma_1(2)$ and $\sigma_1(3)$ lie in $(-1, 1)$ and are nonintegral, and $g_4 = 2 \neq 4 = g_3$.*

Proof. Take $L = 40$ and $Q = 2^{40} = 1099511627776$. Exact integer arithmetic over the first 46 primes gives

N	$QF_{1,N,40}$	$QE_{1,N,40}$
2	−662838684750	11764181250
3	873345886050	12805761250

In each row $|QF| + QE < Q$, which puts the whole certified interval inside $(-1, 1)$, and QE is smaller than the distance from QF to the nearest multiple of Q, which excludes every integer. The margins are 424908761776 and 213359980476 respectively. These are strict integer comparisons rather than inferences from rounded numerical tails. Appendix B replays them. ☐

This witness proves the local condition at one pair of indices. The quantifiers over every h and every cutoff remain.

5 Two denominator floors

The same polynomial bound yields an exact rational enclosure of Π, and any such enclosure excludes an initial range of denominators.

Theorem 5.1 (kernel-decided denominator floor). *Let $a \in \mathbb{Z}$ and $b \in \mathbb{N}_{>0}$. If $\Pi = a/b$ then $b \geq 2^{589} > 10^{177}$, and the same floor holds for every rational equal to S.*

Proof. Put $c = 1229$, the number of primes below 10^4, and $A = \sum_{i=0}^{c-1} p_i 2^{c-i-1}$, so that $A/2^c$ is the cth partial sum of Π. All terms are positive, so $A/2^c \le \Pi$. For the upper bound, $p_{c+j} \le 1250(c+j+1)^4$ and $(c+1+j)^4 \le (c+1)^4(3/2)^j$ for $j \ge 0$, valid because $c \ge 9$; summing the resulting geometric tail with ratio $3/4$ gives

$$\frac{A}{2^c} \le \Pi \le \frac{2A + 5000(c+1)^4}{2^{c+1}}.$$

The four positive integers u, v, u', v' printed in Appendix B satisfy

$$u'v - uv' = 1, \qquad \frac{u}{v} < \frac{A}{2^c}, \qquad \frac{2A + 5000(c+1)^4}{2^{c+1}} < \frac{u'}{v'}, \qquad v + v' \ge 2^{589},$$

all four being integer comparisons after clearing the positive denominators. If $\Pi = a/b$ then $av - bu \ge 1$ and $bu' - av' \ge 1$, and the determinant identity gives

$$b = b(u'v - uv') = (av - bu)v' + (bu' - av')v \ge v + v' \ge 2^{589}.$$

Since $589 \log_{10} 2 > 177$, the floor exceeds 10^{177}. If $S = a/b$, apply the same argument to $\Pi = (a + 2b)/b$. □

The Lean kernel decides the four inequalities on the same literals by `decide +kernel` with no `native_decide`, over a trial-division sieve it re-runs on $[2, 10^4)$: the certificate, the floor for the prime series, and the floor for the gap series. Its analytic input is the same enclosure, over the polynomial bound at line 360. The four Farey literals have 175 to 181 digits.

Theorem 5.2 (certified continued-fraction exclusion)**.** *Every rational equal to Π, and hence every rational equal to S, has reduced denominator $q \ge 2^{39997}$, and therefore $q > 10^{12040}$.*

Evidence. This is an exact finite computation and no Lean declaration carries it. The continued-fraction expansion of Π is developed to 23369 partial quotients at a working scale of 80000 bits. Each quotient is forced by a rational bracket of width 1 in those scaled integer units around the true value, and the separation $|q_n\Pi - p_n| > 0$ is verified against that bracket, so the prefix is the true prefix. No floating-point reconstruction enters it. The bracket uses the Lean-checked tail estimate $p_n \le 1250(n+1)^4$, so no conjecture enters. The classical theorem that a best approximation of the second kind is a convergent then converts the certified prefix into the denominator bound [7]. The receipt is `certified-cf.json`, which records the generating program and its digest, the power-of-two exponent 39997, the bit length 39998, the strict decimal power 12040, the decimal digit count 12041, the largest partial quotient 973919 at index 16442, and an arithmetic self-check of the same routine against the known expansions of e and π. The certified statement is $q \ge 2^{39997}$ and $q > 10^{12040}$. Printing the bit length or the digit count as an exponent overstates the bound by one power of two and one decimal order.

Each floor is finite, and extending either one raises the excluded range and reproduces the same statement form. Neither decides the problem.

6 What cannot supply the missing input

The counterexamples in this section test which information about the gaps could force irrationality. They preserve selected growth, residue and nonconcentration properties while changing the dyadic value. The sparsity result has a different role: it rules out a positive-proportion lower bound for the two-window event, while leaving sparse-witness arguments available.

6.1 Bounded residue-preserving perturbations

Theorem 6.1 (bounded-perturbation obstruction)**.** *Let a_n be natural numbers with $\sum_{n\geq 0} a_n 2^{-(n+1)}$ convergent. For every integer $M \geq 1$ and every cutoff K there are digits $\varepsilon_n \in \{0,1\}$, zero for $n < K$, such that $\sum_{n\geq 0}(a_n + M\varepsilon_n)2^{-(n+1)}$ is rational.*

Proof. Write $A = \sum_{n\geq 0} a_n 2^{-(n+1)}$ and choose a rational $r \in (A, A + M2^{-K})$. A binary expansion of $(r - A)/M$ has the form $\sum_{n\geq K} \varepsilon_n 2^{-(n+1)}$ with $\varepsilon_n \in \{0,1\}$; set $\varepsilon_n = 0$ for $n < K$. The perturbed series converges and equals r. □

The construction is kernel checked. The invariant-property consequence—a property shared by every admissible perturbation cannot by itself force irrationality—is the ordinary logical reading of that construction, rather than a separately named Lean theorem. The construction is also checked at the actual consecutive prime gaps for every $M \geq 1$ and every K, the rational perturbed prime-gap series, with the perturbed gap lying in $[g_n, g_n + M]$ and congruent to g_n modulo M, the gap bounds. Consequently no theorem about the size or the residue of prime gaps that survives such a perturbation can suffice for Problem #251; that class includes the bounded-gap and large-gap theorems cited in Section 8 and equidistribution of p_n modulo any fixed q, on taking $M = 2q$. The perturbed digits need not remain prime gaps.

6.2 Paired corrections and cumulative congruences

The sparse construction preserves cumulative residues as well as coefficient residues. Two adjacent corrections provide a free weighted digit while fixing their unweighted total. A third coordinate is needed to repair the cumulative residue before that pair. We account separately for its location, size and contribution to the series.

Theorem 6.2 (sparse rationalisation with cumulative congruences)**.** *Let $a_n \in \mathbb{N}$ and $A = \sum_{n\geq 0} a_n 2^{-n-1} < \infty$. For every cutoff K and every $f : \mathbb{N} \to \mathbb{R}$ tending to infinity, there are a set $S \subseteq [K, \infty)$ of upper Banach density zero and a nondegenerate interval $I \subset (A, \infty)$ with the following property. Every $r \in I$ has the form*

$$r = \sum_{n\geq 0} (a_n + e_n)2^{-n-1},$$

where $e_n \in \mathbb{N}$ is supported on S, $e_n \leq f(n)$ eventually, and, for every integer $q \geq 1$, there is an N_q such that

$$q \mid e_n \quad \text{and} \quad q \mid \sum_{i<n} e_i \qquad (n \geq N_q).$$

For every $0 < \varepsilon < 1$, the construction can instead be chosen with $e_n \le (\log(n+3))^{\varepsilon}$ eventually and $|S \cap [X, 2X)| = O_{\varepsilon}(X/\log\log X)$. In that case the empirical distributions of original and corrected unnormalised blocks of length $m(X) = o(\log\log X)$, sampled at the same integer starts in $[X, 2X)$, have total variation distance tending to zero.

This theorem has an ordinary proof below. The general interval and pair algebra is covered by the elementary Lean core; the explicit schedule and its density and size estimates are not claimed to be kernel-checked.

Proof. Choose increasing centres n_j, indexed by $j \ge 0$, together with an initial $n_{-1} \ge K$. Their spacings $s_j = n_j - n_{j-1}$ will be at least 4. Let M_j be positive integer moduli with $M_j \mid M_{j+1}$, and set $D_j = 2^{s_j} - 1$. These parameters will be specified below independently of all free digits.

Place the buffer before the pair. Set $C_0 = 0$ and recursively define the deterministic integers

$$c_j = (-C_j) \bmod M_j, \qquad C_{j+1} = C_j + c_j + M_j D_j.$$

For any choice $d_j \in \{0, \dots, D_j\}$, put

$$e_{n_j-1} = c_j, \qquad e_{n_j} = M_j d_j, \qquad e_{n_j+1} = M_j(D_j - d_j), \tag{2}$$

and put $e_n = 0$ off these triples. The spacing makes the triples disjoint. Their total is $c_j + M_j D_j$, independent of d_j, so C_j really is the cumulative correction $C_j = \sum_{i<n_j-1} e_i$ before its buffer, for every choice of free digits. The pair alone could not repair that cumulative residue: its total is already a multiple of M_j. The buffer does, since $M_j \mid C_j + c_j$.

For example, $M = 2, D = 3$ permits the pairs $(0,6), (2,4), (4,2), (6,0)$. Each has total 6, but their weighted contributions have numerators $6, 8, 10, 12$ over 2^{n+2}. A preceding cumulative total $C = 1$ needs the additional buffer $c = 1$ at $n-1$; its weighted contribution is 2^{-n}. This fixed contribution changes the location of the attainable interval, not its free-digit capacity.

Check every late coordinate. Assume that each fixed q divides M_j eventually, and choose J with $q \mid M_J$. At index n_J, just after its buffer, the cumulative sum is divisible by q. Both entries of the pair are also divisible by q, so every later cumulative sum through that stage remains divisible by q. Inductively $q \mid C_j$ and $q \mid M_j$ imply $q \mid c_j$, since $M_j \mid C_j + c_j$. Thus every subsequent buffer, pair entry and zero entry is divisible by q, as is every intermediate cumulative sum. We may take $N_q = n_J$. The first buffer need not itself be divisible by q, and its coordinate $n_J - 1$ is deliberately outside this eventual range.

Choose a schedule for arbitrary f. Define

$$h(n) = \inf_{m \ge n} \min(f(m), m).$$

This finite real number is nondecreasing in n, satisfies $h(n) \le n$, and tends to infinity. Choose $n_{-1} \ge K$ with $h(n_{-1}) \ge 32$ and recursively set

$$\begin{aligned} k_j &= \max\{k \in \mathbb{N} : k \ge 2,\ k!2^{k+2} \le h(n_{j-1})\}, \\ M_j &= k_j!, \qquad s_j = k_j + 2, \qquad n_j = n_{j-1} + s_j. \end{aligned} \tag{3}$$

The maximum exists: $k = 2$ is eligible and $k!2^{k+2}$ tends to infinity. Since h is nondecreasing, the integers k_j are nondecreasing; since $n_j \to \infty$, they tend to infinity. The factorials form a divisibility chain and eventually contain every fixed divisor. At each of the three coordinates n in stage j,

$$0 \le e_n \le M_j 2^{s_j} \le h(n_{j-1}) \le f(n), \qquad e_n \le n_{j-1} < n.$$

Here the buffer is smaller than M_j and each pair entry is at most $M_j D_j$. Thus every correction series converges by comparison with $\sum_n n2^{-n-1}$.

Let $S = \bigcup_j \{n_j - 1, n_j, n_j + 1\}$. It is independent of the digits. To see upper Banach density zero, fix $H > 0$. Outside a finite initial segment, consecutive centres are at least H apart, since $s_j \to \infty$. An interval of length L therefore meets at most $3(L/H + 2)$ late support coordinates, plus the fixed finite number of early coordinates. Divide by L, take the supremum over interval positions, then let $L \to \infty$ and $H \to \infty$.

Retain an interval of values. Put $w_j = M_j 2^{-n_j-2}$ and

$$F_j = \sum_{i \ge j} D_i w_i, \qquad \beta = \sum_{j \ge 0} (c_j 2^{-n_j} + D_j w_j).$$

The size estimates just proved give convergence of both series, $F_j \to 0$, and $F_0 > 0$. The baseline β is positive and independent of the digits. The complete weighted correction, including every buffer, is exactly

$$\sum_n e_n 2^{-n-1} = \beta + \sum_j d_j w_j. \tag{4}$$

For $i > j$, we have $M_i \ge M_j$ and

$$D_i w_i = M_i (2^{-n_{i-1}-2} - 2^{-n_i-2}) \ge M_j (2^{-n_{i-1}-2} - 2^{-n_i-2}).$$

Telescoping over $i = j+1, \dots, J$ and letting $J \to \infty$ yields $F_{j+1} \ge w_j$. This is the needed overlap condition. The intervals $[dw_j, dw_j + F_{j+1}]$, for $0 \le d \le D_j$, cover $[0, F_j]$. For any $y \in [0, F_0]$, choose successive digits leaving each remainder in $[0, F_{j+1}]$. Since these capacities tend to zero, the resulting series has sum y. Equation (4) therefore realises every r in the fixed interval $I = (A + \beta, A + \beta + F_0) \subset (A, \infty)$.

Choose a polylogarithmic schedule. Fix $0 < \varepsilon < 1$. Replace (3) by

$$\begin{aligned} s_j &= \left\lfloor \frac{\varepsilon}{2} \log_2 \log(n_{j-1} + 3) \right\rfloor, \qquad n_j = n_{j-1} + s_j, \\ k_j &= \max\{k \in \mathbb{N} : k \ge 2,\ k! \le (\log(n_{j-1} + 3))^{\varepsilon/4}\}, \qquad M_j = k_j!. \end{aligned} \tag{5}$$

Take the initial cutoff so large that $s_j \ge 4$ and the maximum is nonempty from the first step. Again k_j is nondecreasing and tends to infinity. Keep $D_j = 2^{s_j} - 1$ and the same buffer recursion. All triple entries obey

$$e_n \le M_j 2^{s_j} \le (\log(n_{j-1} + 3))^{3\varepsilon/4} \le (\log(n + 3))^{\varepsilon}.$$

These polylogarithmic bounds still give convergence of F_j and β. Every argument above concerning congruences and interval filling remains valid. Also s_j is comparable, with constants depending on ε, to $\log\log(n_{j-1}+3)$, and $s_j = o(n_{j-1})$. Hence consecutive centres near X are separated by at least a constant multiple of $\log\log X$, giving $|S \cap [X,2X)| = O_\varepsilon(X/\log\log X)$. The same estimate holds on $[X,3X)$.

Finally sample both length-m blocks at the same uniform integer start in $[X,2X)$. They can differ only if that block meets S. Each changed coordinate belongs to at most m such blocks, so the total variation distance is at most

$$\frac{m\,|S \cap [X,2X+m)|}{X} = O_\varepsilon\left(\frac{m}{\log\log X}\right).$$

For integer $m = m(X) = o(\log\log X)$ this tends to zero. This comparison uses the literal, unnormalised blocks and requires no limiting distribution for the original sequence. □

The pair identities, modular repair, and conditional interval filling are the elementary core in `lean/ErdosProblems/Erdos251/SparseRationalisationCore.lean`. The source does not itself establish the schedules (3)–(5), upper Banach density, or block-law transfer. Those parts are the ordinary proof above, not a new Lean or Comparator verification claim. Applied to prime gaps, the construction makes no assertion that the corrected cumulative positions remain prime.

6.3 Algebraic nonconcentration survives the rationalising perturbation

Fixed-block polynomial nonconcentration is a finer property than size and residue, but it too survives the bounded perturbation. The relevant input is Schlage-Puchta's Lemma 4: for every polynomial $F \in \mathbb{Z}[x_0,\dots,x_k]$ which does not vanish identically, $F(g_n,\dots,g_{n+k}) \neq 0$ for almost all n [14, Lemma 4, pp. 5–6]. Say that an integer sequence a has *fixed-block nonconcentration* when it satisfies that conclusion for every $k \geq 0$ and every nonzero $F \in \mathbb{Z}[x_0,\dots,x_k]$. The property transfers across bounded perturbations with no independence or randomness assumption.

Proposition 6.3 (finite counting for shifted gap differences)**.** *Write p_n for the primes indexed from $p_0 = 2$ and $g_n = p_{n+1} - p_n$. For $h \geq 2$ and $r \in \mathbb{Z}$, let $M_{h,r}(N)$ count $n < N$ with $g_{n+h} - g_n = r$. Let $Q_{N,H,r}$ count triples (x,d,s) with $x < p_N$, $0 < d < s \leq H$, $d + r > 0$, and all four integers $x, x+d, x+s, x+s+d+r$ prime. Then, for every $N, H \geq 0$,*

$$(H+1)M_{h,r}(N) \leq (h+1)p_{N+h+1} + (H+1)Q_{N,H,r}.$$

Proof. Windows of total span at most H inject into these four-prime configurations; the sum of all length-$(h+1)$ spans is at most $(h+1)p_{N+h+1}$, which bounds the number of larger spans. □

The finite shifted-count bound is unconditional. To deduce zero density one still needs, for every $\varepsilon > 0$ and all large N, a choice of H making the right side less than $\varepsilon N(H+1)$. The adjacent case $h = 1$ instead requires the corresponding three-prime estimate, because two of the four positions would coincide. The all-shift implication records these analytic premises explicitly; it supplies neither estimate and does not prove irrationality.

Theorem 6.4 (nonconcentration is perturbation-stable)**.** *Let $a : \mathbb{N} \to \mathbb{Z}$ have fixed-block nonconcentration, let $E \subset \mathbb{Z}$ be finite, and let $b_n = a_n + e_n$ with $e_n \in E$ for every n. Then b has fixed-block nonconcentration.*

Proof. Fix $k \geq 0$ and a nonzero $F \in \mathbb{Z}[x_0, \dots, x_k]$. For a tuple $\mathbf{e} = (e^{(0)}, \dots, e^{(k)}) \in E^{k+1}$ put $F_{\mathbf{e}}(x_0, \dots, x_k) = F(x_0 + e^{(0)}, \dots, x_k + e^{(k)})$. The substitution $x_i \mapsto x_i + e^{(i)}$ is a ring automorphism of $\mathbb{Z}[x_0, \dots, x_k]$ with inverse $x_i \mapsto x_i - e^{(i)}$, so $F_{\mathbf{e}} \neq 0$. If $F(b_n, \dots, b_{n+k}) = 0$ then $F_{\mathbf{e}}(a_n, \dots, a_{n+k}) = 0$ for the tuple $\mathbf{e} = (e_n, \dots, e_{n+k})$, whence

$$\{ n : F(b_n, \dots, b_{n+k}) = 0 \} \subseteq \bigcup_{\mathbf{e} \in E^{k+1}} \{ n : F_{\mathbf{e}}(a_n, \dots, a_{n+k}) = 0 \}.$$

Each set on the right has density zero by hypothesis, and the union is over the finite index set E^{k+1}, so a finite union of density-zero sets has density zero. □

Corollary 6.5 (nonconcentration does not force irrationality)**.** *Fix $M \geq 1$ and $K \geq 0$, and let b be the perturbed sequence supplied by Theorem 6.1 at the actual prime gaps. Then $\sum_{n \geq 0} b_n 2^{-(n+1)}$ is rational, $b_n = g_n$ for $n < K$, $b_n - g_n \in \{0, M\}$ and $b_n \equiv g_n \pmod{M}$ for every n, b has fixed-block nonconcentration, and the cumulative sequence $P_n = 2 + \sum_{i<n} b_i$ satisfies $p_n \leq P_n \leq p_n + Mn$ and hence $P_n \sim n \log n$.*

Proof. Everything except nonconcentration is Theorem 6.1 together with $\sum_{i<n} g_i = p_n - 2$. The perturbation takes values in the fixed two-element set $E = \{0, M\}$, so Theorem 6.4 applies to the actual gaps, which have fixed-block nonconcentration by Schlage-Puchta's Lemma 4. □

Fixed-block polynomial nonconcentration, taken together with a prescribed finite prefix of actual gaps, a pointwise bound $b_n \leq g_n + M$, every residue modulo M, positivity and cumulative growth at the prime-number-theorem scale, is therefore compatible with a rational dyadic value. Taking M even also preserves eventual evenness. To preserve a prescribed modulus q together with parity, take $M = 2q$, with the corresponding pointwise bound $b_n \leq g_n + 2q$. The boundary is exact: the terms P_n are not asserted to be prime, and the conclusion says nothing about constraints with a block length or a polynomial family growing with the index.

6.4 The two-window event has density zero

The same lemma bounds the event that the local certificate consumes.

Theorem 6.6 (sparsity of the two-window event)**.** *Fix $h \geq 1$. The set of $N \geq 1$ at which the three hypotheses of Theorem 4.1 hold for the actual prime gaps has density zero. For the same h, the set of N with $g_{N+h+1} = g_{N+1}$ also has density zero.*

Proof. For $N \geq 1$ every gap involved is even, so $\delta_N = g_{N+h+1} - g_{N+1}$ is even. If $|\sigma_h(N)| < 1$, $|\sigma_h(N+1)| < 1$ and $\delta_N \neq 0$, then $\delta_N = 2\sigma_h(N) - \sigma_h(N+1)$ lies in $(-3, 3)$, so $\delta_N = \pm 2$. Apply Schlage-Puchta's Lemma 4 at index $n = N + 1$ with $k = h$ to the two polynomials $F_{\pm}(x_0, \dots, x_h) = x_h - x_0 \mp 2$, neither of which vanishes identically: each of $\{N : \delta_N = 2\}$ and $\{N : \delta_N = -2\}$ has density zero, and the event is contained in their union. The second assertion is the same lemma applied to $F(x_0, \dots, x_h) = x_h - x_0$. □

The two halves point in opposite directions and both are useful. The mismatch hypothesis on its own holds for almost all N, which is strictly stronger than the cofinal statement Proposition 4.3 supplies. The full conjunction is confined to a set of density zero, because the two window conditions force the mismatch to be exactly ± 2. A producer for Problem 8.2 therefore has to be a cofinality statement about a sparse set. Consequently, this event cannot be supplied on a set of positive lower density. A sufficient producer would have to yield cofinally many witnesses in a density-zero set; density zero alone does not exclude an averaging argument capable of detecting such sparse witnesses. The measurement reported in Section 7 is consistent with this: the observed density at $h = 1$ declines like $1/\log p$, and the observed count up to X grows like $X/(\log X)^2$.

6.5 Recurring gap values differing by two do not suffice

Theorem 4.4 reduces the producer to a ± 2 mismatch together with a half-window condition. It is tempting to hope that the mismatch half is the substance, and that two even values differing by 2, each occurring infinitely often as a consecutive prime gap inside a common index residue class, would suffice. At the level of integer-digit recurrences that implication is false, and it stays false under growth at the scale the primes actually have.

Theorem 6.7 (recurring values are not enough)**.** *There is a sequence $(a_n)_{n\geq 1}$ of positive even integers with the following properties. The values 2 and 4 each occur infinitely often at indices divisible by every fixed $t \geq 1$. The sequence is unbounded, not eventually periodic, and satisfies $a_n = O(\log n)$. The series $\sum_{n\geq 1} a_n 2^{-n}$ equals 6, and every scaled tail $\sum_{j\geq 1} a_{N+j} 2^{-j}$ is an integer, so every tail shift is integral. The increasing odd sequence $P_n = 3 + \sum_{j\leq n} a_j$ satisfies $P_n \sim n \log n$.*

Proof. Let logarithms be natural and put $b_n = 2\lceil \frac{1}{2}\log(n+64)\rceil$, so that b_n is even, $b_n \geq 6$, and $b_n = \log(n+64) + O(1)$. Since $\frac{1}{2}\log(n+65) - \frac{1}{2}\log(n+63) < 1$ for every $n \geq 1$, consecutive increments of b lie in $\{0, 2\}$ and $b_{n+1} - b_{n-1} \leq 2$. Call an index n *special* when $n = k!$ or $n = 2k!$ for some $k \geq 5$, and define

$$U_n = \begin{cases} 2b_{n-1} - 2, & n = k!,\ k \geq 5, \\ 2b_{n-1} - 4, & n = 2k!,\ k \geq 5, \\ b_n, & \text{otherwise,} \end{cases} \qquad a_n = 2U_{n-1} - U_n \quad (n \geq 1).$$

The special indices are distinct and never adjacent, since $k!$ and $2k!$ are even and differ by more than one from each other and from the neighbouring special indices. Each U_n is an even integer by construction, so each a_n is an even integer.

At an ordinary index whose predecessor is also ordinary, $a_n = 2b_{n-1} - b_n = b_{n-1} - (b_n - b_{n-1}) \geq 6 - 2 = 4$. At a special index the predecessor is ordinary and $a_n = 2b_{n-1} - (2b_{n-1} - r) = r$, which is 2 at $n = k!$ and 4 at $n = 2k!$. Immediately after a special index carrying r,

$$a_{n+1} = 2(2b_{n-1} - r) - b_{n+1} \geq 4b_{n-1} - 2r - (b_{n-1} + 2) = 3b_{n-1} - 2r - 2 \geq 8.$$

Every a_n is therefore a positive even integer, and $U_n = O(\log n)$ gives $a_n = O(\log n)$. At ordinary indices with ordinary predecessors $a_n \geq b_{n-1} - 2 \to \infty$, so (a_n) is unbounded and hence not eventually periodic.

The definition $a_n = 2U_{n-1} - U_n$ is exactly $a_n 2^{-n} = U_{n-1}2^{-(n-1)} - U_n 2^{-n}$, so for every $N \geq 0$ and $m \geq 1$

$$\sum_{j=1}^{m} \frac{a_{N+j}}{2^j} = U_N - \frac{U_{N+m}}{2^m}.$$

Since $U_n = O(\log n)$ the endpoint tends to zero, so the tail at N equals the integer U_N; at $N = 0$ the series equals $U_0 = b_0 = 6$. Every difference $U_{N+h} - U_N$ is an integer, so every tail shift is integral.

For each t and every sufficiently large k we have $t \mid k!$, hence $t \mid 2k!$, and $a_{k!} = 2$ while $a_{2k!} = 4$: both values recur infinitely often at indices congruent to 0 modulo t.

Finally, summing $a_j = 2U_{j-1} - U_j$ gives $\sum_{j=1}^{n} a_j = 2U_0 + \sum_{j=1}^{n-1} U_j - U_n$. The baseline $\sum_{j<n} b_j = n \log n + O(n)$. The special indices up to n number $O(\log n / \log\log n)$ and each changes the summand by $O(\log n)$, so their total contribution is $O((\log n)^2) = o(n)$. Hence $P_n = 3 + \sum_{j \leq n} a_j \sim n \log n$, and P_n is odd and increasing. □

The factorial schedule above proves recurrence in the zero residue class. The following separate construction strengthens this to every residue class.

6.6 A complete countermodel in every residue class

There is a synthetic integer sequence $(a_n)_{n\geq 1}$ with all of the following properties. Each a_n is positive and divisible by 2, and

$$a_n \leq 4\log(n+1) + 24.$$

For every $t \geq 1$, every $0 \leq r < t$, and every cutoff N, there are $i, j \geq N$ such that

$$i \equiv j \equiv r \pmod{t}, \qquad a_i = 2, \quad a_j = 4.$$

For every integer B and cutoff N, some $n \geq N$ satisfies $a_n > B$. For every $h \geq 1$ there is no cutoff after which $a_{n+h} = a_n$ for all n. Nevertheless, all the complete tails

$$U_N = \sum_{j\geq 1} \frac{a_{N+j}}{2^j} \quad (N \geq 0)$$

are integers, $U_0 = 6$, and $U_{N+h} - U_N \in \mathbb{Z}$ for every $N, h \geq 0$. The sequence

$$P_N = 3 + \sum_{j=1}^{N} a_j$$

is strictly increasing and odd, with $P_N/(N \log N) \to 1$. These are synthetic positions; no P_N is asserted to be prime.

To obtain every residue class, enumerate triples consisting of a positive modulus, a residue, and a repetition index. Choose the corresponding centres c_k in the prescribed classes with $c_0 \geq 100$, $c_{k+1} \geq c_k + 3$ and $c_k \geq 2^{k^2}$ for $k \geq 1$. The parity of the repetition index selects the desired value $v_k = 2$ or 4. Use the even baseline

$$b_n = 6 + 2\left\lfloor \frac{\log(n+1)}{2} \right\rfloor, \qquad U_n = \begin{cases} 2b_{n-1} - v_k, & n = c_k, \\ b_n, & n \notin \{c_k : k \geq 0\}, \end{cases}$$

and define $a_n = 2U_{n-1} - U_n$. The separation of the centres and the baseline increment bound give positivity and the displayed logarithmic bound. Finite telescoping leaves $U_{N+m}/2^m$, which tends to zero; thus these carries are the complete tails. The sparse changes to the baseline have negligible contribution after division by $N \log N$, which gives the stated growth of P_N.

The full statement is kernel-checked in Lean.

This rules out an inference from these coefficient properties alone to nonintegral tail shifts. It gives no counterexample to the prime-gap problem and does not reproduce the extreme large gaps of the primes.

The consequence is exact. Positivity, evenness, unboundedness, non-eventual periodicity, a pointwise logarithmic bound, cumulative growth at the prime-number-theorem scale, and the recurrence of two even values differing by 2 inside every index residue class are jointly compatible with an integral tail at every index. Any route through that input must use a property of the consecutive primes beyond this list. The sequence (a_n) is synthetic and the P_n are not asserted to be prime. The pointwise logarithmic bound also precludes the known large-gap behaviour, so the theorem does not speak to hypotheses that use extreme gaps.

6.7 Two further boundaries

Proposition 6.8 (quadratic polynomial-shift countermodel)**.** *Put $c_n = 2(n^2+4n+2)$ and $U_n = 2(n+4)^2$. Then c_n is positive, even and strictly increasing, $U_{n+1} = 2U_n - c_{n+1}$, every shift $U_{N+h} - U_N$ is integral, $c_{n+1} - c_n = 4n + 10$ is never ± 2, and*

$$\sum_{j\geq 1} \frac{c_j}{2^j} = 32.$$

Proof. Direct expansion gives the recurrence, and the finite telescope is $\sum_{j=1}^{n} c_j 2^{-j} = 32 - 2(n+4)^2 2^{-n}$, whose last term tends to zero. The remaining assertions follow from the integer values of U_n and from $c_{n+1} - c_n = 4n + 10 \geq 10$. □

The series is the one starting at $j = 1$, whose initial tail state is $U_0 = 32$; under the opening convention $\sum_{n\geq 0} c_n 2^{-(n+1)}$ the same word has value 18, and the two ranges must be kept apart. The formal construction checks the recurrence, strict growth, integrality of every shift, exclusion of adjacent differences of size two, and the value 32 for the shifted series, whose summand is $c_{n+1}/2^{n+1}$; the rational value is recorded at line 109. Positivity, parity, strict growth, unboundedness and nonperiodicity are therefore jointly compatible with rationality, and the adjacent trigger of Theorem 4.4 fails at every index. The telescoping mechanism is the one Kovač used against the variable-denominator conjecture printed under the problem [18, Theorem 1 and proof, pp. 1–2]; the word carries no priority claim and is not a result about Problem #251.

Rationality also fails to force the coefficients to repeat. Let $K : \mathbb{N} \to \mathbb{Q}$ be arbitrary and put $\kappa_n = 2K_n - K_{n+1}$: the carry coefficient.

Proposition 6.9 (exact telescoping)**.** *For every $n \geq 0$, $\sum_{i=0}^{n-1} \kappa_i 2^{-(i+1)} = K_0 - K_n 2^{-n}$.*

Proof. A routine induction; the added term is $(2K_n - K_{n+1})2^{-(n+1)} = K_n 2^{-n} - K_{n+1} 2^{-(n+1)}$. □

Formalised as the carry telescoping identity. Taking $K_0 = \frac{5}{2}$ and $K_n = 2n + 2$ for $n \geq 1$ gives $\kappa_0 = 1$ and $\kappa_n = 2n$, so the coefficients $1, 2, 4, 6, 8, \ldots$ are positive, even after the first term, unbounded and not eventually periodic, while their dyadic sum is $\frac{5}{2}$. Rationality alone therefore cannot imply eventual periodicity even for a positive, parity-correct integer coefficient sequence, and no argument may play periodicity of a rational value against Proposition 4.3.

Finally, two escape tests that look like independent producers are equivalent to the conclusion they were introduced to supply. For fixed h write $\delta_N = g_{N+h+1} - g_{N+1}$, $D_N = \sigma_h(N)$ and $B_{h,N,r} = \sum_{i=0}^{r-1} 2^{r-1-i}\delta_{N+i}$, so that $D_{N+r} = 2^r D_N - B_{h,N,r}$, and call the data-dependent affine condition $\mathcal{A}_{N,r}$ the assertion $D_{N+r} \in -B_{h,N,r} + 2^{r+1}\mathbb{Z}$.

Theorem 6.10 (affine and fixed-lattice circularity)**.** *For every rational dyadic tail recurrence and all $h, N, r \geq 0$, $\mathcal{A}_{N,r}$ holds if and only if $D_N \in 2\mathbb{Z}$. Consequently, if every δ_N is even, cofinal failure of $\mathcal{A}$ is equivalent to nonintegrality of D_N for arbitrarily large N. The same equivalence holds for the fixed-lattice replacement, under any bound $|D_N| \leq b(N)$ satisfying $2b(N+r)q < 2^r$ for some r at every N and every positive integer q.*

Proof. Substituting $D_{N+r} = 2^r D_N - B_{h,N,r}$ into $\mathcal{A}_{N,r}$ cancels the observed block and leaves $D_N = 2z$. After one recurrence step, evenness of δ_N identifies even integrality at $N + 1$ with ordinary integrality at N; the reverse implication chooses a nonintegral D_N and depth $r = 0$. For the fixed-lattice form, eventual integrality and the block identity give $B_{h,N,r} - 2^r z = -D_{N+r}$, so a strict separation exceeding $b(N + r)$ is contradictory; conversely a nonintegral D_N with reduced denominator q has distance at least $1/q$ from every integer, and scaling that separation by 2^r at a legal depth gives the required strict inequality. □

These are the pointwise affine identity, its cofinal form, and the fixed-lattice equivalence. The apparent affine hierarchy contains no depth-dependent information, and the fixed lattice removes the data dependence at the cost of a growth condition under which rational denominator separation already forces every required escape. Neither is an independent source of information about consecutive primes.

7 What is measured

Two finite computations report on the two producers. Each covers one bounded range and supplies no cofinal statement.

Over the 6 841 648 primes below 1.2×10^8 and offsets $h = 1, \ldots, 16$, the reduced two-window event of Theorem 4.4 occurs at every offset, with density between 0.00418 and 0.008248 and the two signs near balanced; at $h = 1$ there are 56 427 occurrences among 6 841 564 candidate indices, the last at the prime 119 995 753. Rescaled by $\log p$ the band densities run from 0.17671 down to 0.13148 with decelerating drift over the measured bands. This finite observation does not establish an asymptotic counting law or cofinality. The tails in this scan are double-precision floating-point, so an individual hit is a numerical observation and carries no certificate; the median distance from the shift to the nearer window boundary is 0.127 and the worst case over 34 000 hits is 6.1×10^{-7}, eight orders above the working resolution, and 3 200 sampled hits across $h = 1, \ldots, 8$

were rechecked against the literal three-part statement with no violations. Proposition 4.6 is the one pair certified by exact integer arithmetic. The receipt is `adjacent-mismatch.json`.

Over the 1 270 607 primes below 2×10^7, and for every modulus $t \leq 20$ and every residue class modulo t, the free-pair event of Theorem 3.4 has witnesses running to the last usable index: the leanest class at $t = 20$ carries 90 150 witnesses and the latest witness sits at index 1 270 520 of 1 270 540. The recorded floating-point witnesses pass the stronger two-window test over this finite range; they are numerical observations, not exact certificates or a proof of either cofinal producer. What the scan does not supply is uniformity in t and in the cutoff. The receipt is `free-pair.json`. The public computation guide provides the programs, dependencies and exact replay commands for all three recorded runs, including the continued-fraction calculation above.

8 The remaining obligation

Problem #251 is open. The exact unresolved condition, equivalent to irrationality by Theorem 3.3 and the escape equivalence for the actual tail, is the following.

Problem 8.1 (universal prime-gap shift escape)**.** For every $h \geq 1$ and every N_0, prove that some $N \geq N_0$ satisfies

$$\sum_{j \geq 1} \frac{g_{N+h+j} - g_{N+j}}{2^j} \notin \mathbb{Z}. \tag{8.1}$$

Problem 8.2 (cofinal adjacent small mismatch)**.** For every $h \geq 1$ and every N_0, prove that some $N \geq N_0$ satisfies

$$|\sigma_h(N)| < 1, \qquad |\sigma_h(N+1)| < 1, \qquad g_{N+h+1} \neq g_{N+1}. \tag{8.2}$$

Corollary 4.2 makes Problem 8.2 sufficient for Problem 8.1, and Theorem 3.4 supplies an equivalent target in which the offset is free. The three formulations are equivalent or ordered by implication; none of them is an analytic saving on its own.

What the proved obstructions leave. By Theorem 6.1 no hypothesis stable under bounded residue-preserving perturbation can work; by Corollary 6.5 adding fixed-block polynomial nonconcentration to that list does not repair it; the construction in Section 6.6 shows that recurring gap values differing by 2 in every residue class do not work at the level of integer recurrences; and by Theorem 6.6 the event in (8.2) has density zero, so it must be produced cofinally on a sparse set. Isolated small gaps, isolated large gaps and average gap estimates are insufficient. Any prescribed finite prefix can be preserved while the complete sum is rationalised, so that prefix alone cannot force irrationality or a cofinal small-tail producer. A finite block together with a proved tail bound can still certify one window: both inequalities in (8.2) contain the complete infinite continuation, and Proposition 4.5 supplies the corresponding finite reduction. Nor does parity help, since after the first gap every g_n is even.

The finite reduction is available. Proposition 4.5 turns each window condition into an integer comparison at truncation length L, and for fixed h and $\varepsilon > 0$ the choice $L = \lceil (4+\varepsilon)\log_2(N+2) \rceil$ makes $E_{h,N,L} = O_h(N^{-\varepsilon})$. A prescribed positive margin η_h therefore converts (8.2) into a statement about the joint distribution modulo powers of two of the finite block $(g_{N+h+1} - g_{N+1}, \cdots, g_{N+h+L} - g_{N+L})$ over a window of logarithmic length. Deciding one instance costs $O(\log N)$ gaps together with the elementary prime bound; producing them cofinally, with uniformity in h and in the cutoff, is the open work.

The strongest published results on prime gaps address a different shape of question. Zhang's bounded-gap theorem [15, Theorem 1, p. 1122] produces infinitely many bounded consecutive-prime gaps. Maynard bounds $\liminf_n (p_{n+m} - p_n)$ for every fixed m, giving bounded clusters of every fixed size [12, Theorem 1.1, p. 384], with the explicit unconditional bound $\liminf_n (p_{n+1} - p_n) \leq 600$ [12, Theorem 1.3, p. 385]. The large-gap theorem of Ford, Green, Konyagin, Maynard and Tao bounds the largest single consecutive-prime gap below X [6, Theorem 1, p. 66]. A theorem giving bounded clusters at each fixed cluster size does not by itself supply a weighted condition on windows whose length grows with the basepoint, and none of these results is used as a proof input here.

Conditionally the picture is different. Tao's comment names uniform quantitative prime-tuples control of about $\log\log n$ consecutive gaps as the plausible route [17, comment of 7 October 2025], and Land's draft carries that out under Kuperberg's uniform prime-tuples conjecture [9]. The obstructions of Section 6 say which unconditional substitutes cannot replace that hypothesis.

What remains to be formalised. The two finite certificates of Propositions 4.5 and 4.6, the continued-fraction exclusion of Theorem 5.2, the nonconcentration transfer of Theorem 6.4 with its corollary, the sparsity statement of Theorem 6.6, and the construction of Theorem 6.7 are ordinary mathematical proofs with exact arithmetic replay and no Lean declaration. Everything else linked above is checked by the kernel at the pinned commit.

Statements and declarations

Artefact and data availability. The pinned formal-source revision contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. Proof authority rests in those pinned sources. The present text is exposition and navigation.

Lean checks each proof term against the fixed library version, and the sources linked here contain no proof placeholders and no project-defined axioms; Lean does not authorise the exposition, the citation choices, or the interpretation, for which the author remains responsible. The two receipts named in Section 7 and the continued-fraction receipt of Theorem 5.2, together with their generating programs, are public at the immutable revision linked there. The computation guide records dependencies, replay commands, source provenance, and the exact versus floating-point evidence boundary.

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A An elementary polynomial bound for the primes

We prove $p_n \le 1250(n+1)^4$ for every $n \ge 0$. Let $\pi(x)$ count the primes at most x. For an integer $m \ge 4$,

$$4^m < m\binom{2m}{m} \le m(2m)^{\pi(2m)}. \tag{6}$$

For the first inequality, $m\binom{2m}{m}/4^m$ equals $70/64$ at $m = 4$ and its ratio at successive indices is $(2m+1)/(2m) > 1$. For the second, the exponent of a prime ℓ in $\binom{2m}{m}$ is $\sum_{k\ge1}(\lfloor 2m/\ell^k\rfloor - 2\lfloor m/\ell^k\rfloor)$, each summand is 0 or 1, and every summand with $\ell^k > 2m$ vanishes; hence each prime power dividing $\binom{2m}{m}$ is at most $2m$, and there are $\pi(2m)$ of them.

Now fix $n \ge 0$, put $x = n + 5$ and $m = x^4 \ge 625$, and suppose $\pi(2m) \le n$. Since $x \le 2^x$ and $n + 4(n+1)x = 4x^2 - 15x - 5 \le 2x^4$,

$$m(2m)^n = 2^n x^{4(n+1)} \le 2^{n+4(n+1)x} \le 2^{2x^4} = 4^m,$$

contradicting (6). Hence $\pi(2m) > n$, so at least $n+1$ primes lie below $2m$ and therefore

$$p_n \le 2(n+5)^4 \le 1250(n+1)^4,$$

the last step because $(n+5) \le 5(n+1)$ for $n \ge 0$. The same bound is checked by the kernel at line 360.

B Integer certificates

The following calculation uses trial division and integer arithmetic only. It reproduces the two rows of Proposition 4.6 and every inequality used in Theorem 5.1. The four literals are the certificate the Lean kernel decides in `KernelDenominatorFloor.lean`; their use here requires no floating-point approximation. Adjacent quoted strings are concatenated by Python.

```
from math import isqrt

primes = [n for n in range(2, 10000)
          if all(n % d for d in range(2, isqrt(n) + 1))]
gaps = [q - p for p, q in zip(primes, primes[1:])]
P = lambda x: x**4 + 8*x**3 + 36*x*x + 104*x + 150
Q = 2**40
expected = [(-662838684750, 11764181250),
            (873345886050, 12805761250)]
for N, target in zip((2, 3), expected):
    D = sum((gaps[N+1+j] - gaps[N+j])*2**(40-j)
            for j in range(1, 41))
    B = 1250*(P(N+43) + P(N+42))
    distance = min(D % Q, Q - D % Q)
    if (D, B) != target or not (abs(D)+B < Q and B < distance):
        raise ArithmeticError("tail certificate failed")
if gaps[4] == gaps[3]:
    raise ArithmeticError("gap mismatch failed")

u = int(
    "8065641857152652932176019632186898003271162829171466334827"
```

```
    "3083607794415278717445033509407855988903369988525550746159"
    "7355889792250084202344821020139160956663658789718168152662"
    "0217"
)

v = int(
    "2194945124413663232143970924541263312422069524635615360518"
    "4247077351958221816830720189289904831662955084392690248683"
    "1216291723988537733235173040607254496838530213867781442335"
    "1745"
)

up = int(
    "6539437101498162626882411892470905222108260008568555445975"
    "3026163315521784668689909712759876562484620059098138417469"
    "5232839888185316374272333277389611483117334000493867584923"
    "912"
)

vp = int(
    "1779611075789866551198427241621709630120136323281598176637"
    "8490605249229735501968764904036152970729491656650873938580"
    "6203025466313389810291243786005499164537499383612081805160"
    "767"
)

c = len(primes)
A = sum(p*2**(c-1-i) for i, p in enumerate(primes))
checks = (c == 1229,
          u > 0 and v > 0 and up > 0 and vp > 0,
          up*v - u*vp == 1,
          u*2**c < A*v,
          (2*A + 5000*(c+1)**4)*vp < up*2**(c+1),
          v+vp >= 2**589,
          2**589 > 10**177)
if not all(checks):
    raise ArithmeticError("denominator certificate failed")
print("Both tail rows and the denominator floor verified.")
```

C Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision ee650b32b8b2. The summation-by-parts and prime-bound declarations are prime-specific. Section 3 is stated for arbitrary integer digits and arbitrary rational or real orbits, while Section 4 records the actual-gap specialisation. The concrete prime-gap tail, its unconditional convergence, its recurrence and the real-to-rational scaled-tail bridge are defined and checked in `RealPrimeGapTail.lean` and `PrimeGapDyadicTail.lean`; the free-pair lattice and its actual-series equivalence in `FreePairReduction.lean`; the kernel-decided denominator certificate in `KernelDenominatorFloor.lean`; the rational countermodels in `BoundedPerturbationCountermodel.lean` and `PolynomialGapSeriesValue.lean`; and the two circularity theorems in `AffineCylinderCol-`

`lapse.lean`, with the signed normal form in `AffineShiftEscape.lean` and the lcm diagonal in `OrderLatticeDiagonal.lean`.

D Extended record: displaced results, ordinary proofs and corrections

This section holds mathematics that belongs to the complete record for Problem #251 and sits outside the short note. Every statement here is labelled by its evidence class in the same vocabulary the note uses.

D.1 The totient witness and forward propagation

The note states the exact denominator law (3.1) and uses it directly. The classical witness that produced it is worth recording, because it is the form in which the shift length first appears.

Proposition D.1 (a shift of totient length)**.** *Let $T : \mathbb{N} \to \mathbb{Q}$ satisfy the dyadic tail recurrence with integer digits. If the reduced denominator d of T_N is odd, then $\sigma_{\varphi(d)}(N)$ is an integer.*

Proof. Since d is odd, 2 and d are coprime, so Euler's congruence gives $d \mid 2^{\varphi(d)} - 1$. Writing $2^{\varphi(d)} - 1 = dk$ and $T_N = u/d$ in lowest terms, $(2^{\varphi(d)} - 1)T_N = ku$ is an integer, and the integral-shift criterion transfers this to the shift. □

Proposition D.2 (propagation)**.** *If $\sigma_h(N)$ is an integer, then $\sigma_h(N+k)$ is an integer for every $k \geq 0$.*

Proof. The shift step identity gives $\sigma_h(N + 1) = 2\sigma_h(N) - (g_{N+h+1} - g_{N+1})$, an integer combination of an integer and two digits. Induct on k. □

These are the totient shift and the propagation theorems cited in the note. Two worked instances. If $\operatorname{den} T_N = 3$ then $\varphi(3) = 2$ and $3T_N$ is an integer, so $\sigma_2(N)$ is integral while $\sigma_1(N)$ is not; if $\operatorname{den} T_N = 5$ then $\sigma_4(N)$ is integral. In the orbit with every digit zero and $T_0 = 1/12$, the denominator is $12 = 2^2 \cdot 3$, the orbit reaches $T_2 = 1/3$ with odd denominator at $s = 2$, and $h = \varphi(3) = 2$ is exactly the shift length seen to be integral from index 2 onwards. The totient is a witness rather than the classification: the exact criterion is $\operatorname{den}(T_N) \mid 2^h - 1$, and the least admissible h is the multiplicative order of 2 modulo the odd part.

D.2 The finite truncation criterion

The note gives the explicit remainder bound for the actual gaps. The general truncation statement behind it, valid for any dominating majorant, is the following.

Proposition D.3 (finite truncation)**.** *Let $M(n) \geq g_n$ for every n, assume the series below converges, and put*

$$S_{h,N,L} = \sum_{j=1}^{L} \frac{g_{N+h+j} - g_{N+j}}{2^j}, \qquad R_{h,N,L}(M) = \sum_{j>L} \frac{M(N+h+j) + M(N+j)}{2^j}.$$

If for every $h \geq 1$ and every N_0 there are $N \geq N_0$ and $L \geq 1$ with $\operatorname{dist}(S_{h,N,L}, \mathbb{Z}) > R_{h,N,L}(M)$, then the universal shift escape of Problem 8.1 holds.

Proof. The part of $\sum_{j\geq 1}(g_{N+h+j} - g_{N+j})2^{-j}$ omitted from $S_{h,N,L}$ has absolute value at most $R_{h,N,L}(M)$, so under the displayed inequality the full sum lies at positive distance from every integer. □

The truncation is an exact modular small-arc problem. Writing the integral dyadic block

$$D_{h,N,L} = \sum_{j=1}^{L} 2^{L-j}(g_{N+h+j} - g_{N+j}), \qquad S_{h,N,L} = \frac{D_{h,N,L}}{2^L},$$

one has

$$\operatorname{dist}(S_{h,N,L}, \mathbb{Z}) = 2^{-L}\min\{D_{h,N,L} \bmod 2^L,\ 2^L - (D_{h,N,L} \bmod 2^L)\},$$

where $D_{h,N,L} \bmod 2^L$ is the least nonnegative residue, including when $D_{h,N,L} < 0$. The finite criterion therefore asks the residue of $D_{h,N,L}$ to avoid the two arcs of radius $2^L R_{h,N,L}(M)$ around 0 modulo 2^L. A one-block certificate would be a prime-gap theorem producing such an avoided arc on a logarithmic block. This rewriting is paper-level and carries no Lean declaration.

D.3 Divisor-hitting shift escape

Problem D.4 (divisor-hitting shift escape)**.** For every $r \geq 1$, does some positive multiple of r escape cofinally, in the sense of (8.1) at shift length mr for some $m \geq 1$?

This is an exact reformulation, not a weaker mathematical target. If T_0 is irrational, every positive shift is nonintegral by the block identity, so divisor-hitting escape follows with $m = 1$. Conversely, rationality gives a shift $h \geq 1$ that is integral at every sufficiently late index. For each positive integer m, the identity

$$T_{N+mh} - T_N = \sum_{j=0}^{m-1} (T_{N+(j+1)h} - T_{N+jh})$$

makes every multiple mh eventually integral. Taking $r = h$ contradicts divisor-hitting escape. This ordinary deduction uses the checked rationality classification and a finite telescope; rearranging the quantifiers supplies no new estimate for the actual prime gaps.

D.4 Two ordinary proofs about the free-pair producer

Both arguments below are ordinary mathematical deductions. The first uses the prime number theorem, while the second consumes the free-pair lattice proved above. Neither deduction is claimed to be kernel-checked; the exact tail recurrence and free-pair statements they use have their separate linked Lean evidence.

State compression. Exchanging the order of summation gives $\sum_{N\leq X} T_N \leq (p_{X+1} - p_0) + 2T_X$. The prime number theorem [13, Ch. 6] also gives

$$\frac{T_X}{p_X} = \sum_{j\geq 1} \frac{g_{X+j}}{2^j p_X} \longrightarrow 0. \tag{7}$$

Indeed, for each fixed j the summand without 2^{-j} tends to zero because $p_{X+j}/p_X \to 1$. The usual two-sided bounds $p_n \asymp n\log n$ dominate g_{X+j}/p_X by $K(1+j)^2$, independently of X; multiplication by 2^{-j} therefore permits dominated convergence. Applying the first-moment bound at $2X$ and using (7), there is an absolute K_0 such that

$$\sum_{X<N\le 2X} T_N \le K_0 X \log X$$

for all large X. Thus, for each fixed $C>1$, Markov's inequality leaves at least $(1-1/C)X$ indices in $(X,2X]$ with $T_N \le K_0 C\log X$.

Under rationality, take X beyond the point where the reduced denominator is the fixed odd integer d. On this set T_N lies in $d^{-1}\mathbb{Z}_{\ge 0}$ and takes at most $dK_0C\log X+1$ values. Pigeonhole therefore gives one value at $\gg_{C,d} X/\log X$ indices. This compression uses the growth of the actual primes. It is false for a general rational integer-digit orbit: the quadratic countermodel of Proposition 6.8 has the strictly increasing tails $T_N = 2(N+4)^2$. Even for the primes, equal-tail pairs have difference zero and hence do not supply the nonintegral difference required by the free-pair criterion. The naive form $\sum_{N\le X} T_N \le p_{X+1}$ is false.

Redundancy of the gap condition. Under the free-pair lattice the difference $T_M - T_N$ is an integer whenever the modulus divides the offset, and an integer cannot lie in $(\frac{1}{2},1)$. The half-window component of the reduced event of Theorem 4.4 is therefore already a contradiction on its own, and the gap-mismatch condition adds nothing to it. That changes which component a proof has to produce.

D.5 A bounded companion to the recurring-values countermodel

Theorem 6.7 carries a logarithmic growth profile so that its cumulative sequence matches the primes. The mechanism is visible in a much smaller example, which is recorded here because it is checkable by hand.

Proposition D.5 (bounded recurring-values countermodel)**.** *Put $U_0 = 4$ and, for $n\ge 1$, $U_n = 6$ when $n = k!$ for some $k\ge 3$ and $U_n = 4$ otherwise, and set $a_n = 2U_{n-1} - U_n$ for $n\ge 1$. Then $a_n \in \{2,4,8\}$, the value 2 occurs at every index $k!$ and the value 4 at every index $2k!$, so both recur infinitely often at indices divisible by any fixed $t\ge 1$. The series $\sum_{n\ge 1} a_n 2^{-n}$ equals 4 and every tail $\sum_{j\ge 1} a_{N+j}2^{-j}$ equals the integer U_N.*

Proof. For $k\ge 3$ the index $k!$ is even and at least 6, and $k!-1$ is odd, so the predecessor of a spike is an ordinary index; hence $a_{k!} = 8-6 = 2$, $a_{k!+1} = 12-4 = 8$, and $a_n = 8-4 = 4$ elsewhere. For $k\ge 2$ one has $k! < 2k! < (k+1)!$, so $2k!$ is not a factorial, and $2k!-1$ is odd so it is not a factorial either; therefore $a_{2k!} = 4$. Since $t \mid k!$ for every large k, both values recur in the residue class 0 modulo t. The identity $a_n 2^{-n} = U_{n-1}2^{-(n-1)} - U_n 2^{-n}$ telescopes to $\sum_{j=1}^{m} a_{N+j}2^{-j} = U_N - U_{N+m}2^{-m}$, and U is bounded, so the endpoint vanishes. □

The cumulative sequence of this word grows linearly, which is the one property that separates it from the primes and the reason the note carries the logarithmic version instead. Both refute the same generic implication.

D.6 The measured densities in full

The note reports the range of the adjacent-mismatch density. The per-offset figures below come from the same scan over the 6 841 648 primes under 1.2×10^8, with double-precision tails.

h	events	density	$\delta = +2$	$\delta = -2$	last event at prime
1	56 427	0.008248	28 022	28 405	119 995 753
2	31 979	0.004674	15 742	16 237	119 993 807
3	30 233	0.004419	15 093	15 140	119 998 321
4	29 262	0.004277	14 606	14 656	119 993 473

Every offset $h \leq 16$ has events, with minimum density 0.00418 and maximum 0.008248; the two signs are near balanced at every offset. For $h = 1$ the eight band densities, and the same figures after rescaling by the band mean of $\log p$, run

$$0.011285,\ 0.008951,\ 0.008303,\ 0.007936,\ 0.007651,\ 0.007507,\ 0.007253,\ 0.007094;$$

$$0.17671,\ 0.15058,\ 0.14420,\ 0.14067,\ 0.13766,\ 0.13667,\ 0.13333,\ 0.13148.$$

The rescaled drift is 0.744 at $h = 1$ and lies between 0.744 and 0.8121 across all offsets. Theorem 6.6 proves that the limiting density is zero, so the observed decline is the expected behaviour rather than a failure of the event.

D.7 Corrections to earlier records

The superseded exclusion. An earlier internal record narrated a denominator exclusion at 10^{602}. That figure is correct as written and is superseded twice over: by the kernel-decided floor $2^{589} > 10^{177}$ of Theorem 5.1, which is smaller but kernel-checked, and by the certified continued-fraction exclusion $2^{39997} > 10^{12040}$ of Theorem 5.2, which supersedes it in strength. The receipt records the decimal digit count 12041; the certified decimal statement uses the exponent 12040.

The named missing input. An earlier record named Hardy-Littlewood k-tuple correlation of consecutive gaps at a fixed offset as the missing input. With the offset freed by Theorem 3.4 the route needs, for each modulus t, cofinally many congruent pairs with certified nonintegral tail difference. A further record proposed that two even values differing by 2, each occurring infinitely often inside a common index residue class, would supply the two-condition form. No proof of that implication was ever produced, and Theorem 6.7 shows that no proof exists at the level of integer-digit recurrences. That entry is withdrawn.

The formal boundary. An earlier reading of the short note placed the real-tail bridge and the real form of the adjacent-pair consumer on the paper side. Both are kernel-checked, at line 48 and line 99 of the real prime-gap tail module. A second reading placed the kernel-decided denominator floor and the free-pair criterion outside the source revision the note pins. Both modules are present at that revision, and both are linked from the note.

E Complete result-family map

This section places every registered family for this problem in the shared 70-family reader order. The five display bands control exposition only; the separate promotion state currently covers 6 families and is reported but does not hide or strengthen any family. Mathematical statements and evidence modes come from the public claim registry. Across all eight problems the public result-atom catalog contains 681 exact packet coordinates; this problem contributes 30. The recovered source catalog contains 682 rows; 1 row(s) belong to families absent from the current claim registry and remain disclosed as detached source rows. Catalog rows expose bounded statement excerpts plus full-source digests, not a claim that every complete packet statement is reproduced here.

E.1 Prime gap reformulation

Reader position. 4 of 70; display band: front door. Formal editorial disposition: promote. These are separate classifications.

Reader entry. Exact finite summation by parts, the unconditional infinite prime-gap identity, and irrationality equivalence.

Exact finite summation by parts, the unconditional infinite prime-gap identity, and irrationality equivalence.

Authority and reach. Lean kernel; Comparator-selected; locally proved result; novelty unassessed.

Exact boundary. The equivalence does not prove irrationality of either series.

Result-atom population. 4 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. exact selected interface; 1 executable interface(s), 1 repository-registered selected result interface(s).

Publication placement. This family remains in the complete long record and is not a short-note headline.

E.2 Dyadic tail integrality classification

Reader position. 11 of 70; display band: major result. Formal editorial disposition: merge. These are separate classifications.

Reader entry. For any integer-digit dyadic tail recurrence T(N+1) = 2*T(N) - g(N+1), the rational h-shift is integral exactly when the reduced denominator of T(N) divides 2^h - 1, equivalently when 2^h is congruent to 1 modulo that denominator. For the real recurrence, the load-bearing normal form says that irrationality of T(0) is equivalent to every positive tail shift being nonintegral; the eventual-integrality equivalence and denominator/order laws are supporting mechanism evidence.

For any integer-digit dyadic tail recurrence T(N+1) = 2*T(N) - g(N+1), the rational h-shift is integral exactly when the reduced denominator of T(N) divides 2^h - 1, equivalently when 2^h is congruent to 1 modulo that denominator. For the real recurrence, the load-bearing normal form says that irrationality of T(0) is equivalent to every positive tail shift being nonintegral; the eventual-integrality equivalence and denominator/order laws are supporting mechanism evidence.

Authority and reach. Lean kernel; Comparator-selected; abstract Lean-checked classification and normal form; novelty unassessed.

Exact boundary. These exact statements cover arbitrary integer-digit dyadic tail recurrences with the declared rational and real state types; they classify integrality and give an irrationality normal form but do not supply a cofinal nonintegral shift for the consecutive-prime-gap recurrence. The generic hypotheses therefore do not close the prime-specific producer gap: no prime-gap irrationality, novelty, significance, or external review is claimed, and Erdős #251 remains open. This is one abstract classification family, distinct from prime_gap_reformulation, not a duplicate frontier claim.

Result-atom population. 10 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. exact selected interface; 1 executable interface(s), 1 repository-registered selected result interface(s).

Publication placement. This family remains in the complete long record and is not a short-note headline.

- `ErdosProblems.Erdos251.tailShift_integral_iff_den_dvd_mersenne`
- `ErdosProblems.Erdos251.tailShift_integral_iff_two_pow_modEq_one`
- `ErdosProblems.Erdos251.not_irrational_initial_iff_exists_eventually_integral_positive_tailShift`
- `ErdosProblems.Erdos251.irrational_initial_iff_all_positive_tailShifts_nonintegral`

E.3 Integral shift classification

Reader position. 27 of 70; display band: mechanism. Formal editorial disposition: merge. These are separate classifications.

Reader entry. Block identities and exact denominator criteria classify rationality through integral positive tail shifts.

Block identities and exact denominator criteria classify rationality through integral positive tail shifts.

Authority and reach. Lean kernel; locally proved result; novelty unassessed.

Exact boundary. The concrete prime-gap producer remains missing.

Result-atom population. 1 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. not selected for comparator; 0 executable interface(s), 0 repository-registered selected result interface(s).

Publication placement. This family is also admitted to the short note.

- `ErdosProblems.Erdos251.irrational_initial_iff_all_positive_tailShifts_nonintegral`
- `ErdosProblems.Erdos251.not_irrational_initial_iff_exists_eventually_integral_positive_tailShift`
- `ErdosProblems.Erdos251.tailShift_integral_iff_den_dvd_mersenne`

- `ErdosProblems.Erdos251.tailShift_integral_iff_two_pow_modEq_one`

E.4 Small mismatch criterion

Reader position. 45 of 70; display band: frontier. Formal editorial disposition: hold. These are separate classifications.

Reader entry. For the actual prime-gap dyadic tail, cofinally many adjacent pairs of strictly small h-shifts with unequal corresponding prime gaps force failure of eventual RatIntegral h-shifts. The adjacent-pair zero obstruction is the load-bearing mechanism; exact non-eventual periodicity of consecutive prime gaps supplies the actual-prime-gap contrast.

For the actual prime-gap dyadic tail, cofinally many adjacent pairs of strictly small h-shifts with unequal corresponding prime gaps force failure of eventual RatIntegral h-shifts. The adjacent-pair zero obstruction is the load-bearing mechanism; exact non-eventual periodicity of consecutive prime gaps supplies the actual-prime-gap contrast.

Authority and reach. Lean kernel; Comparator-selected; conditional actual-prime-gap endpoint reduction; novelty and significance unassessed.

Exact boundary. The representative assumes $T : \mathbb{N} \to \mathbb{Q}$, the DyadicTailRecurrence for the actual prime-gap digits, and for every N_0 an $N \geq N_0$ whose adjacent h-shifts both lie strictly in (-1, 1) while the corresponding prime gaps differ. Under that cofinal small-mismatch supply it excludes eventual RatIntegral h-shifts; it does not prove the supply or actual smallness. rationalPrimeGapTail_has_positive_shift_not_eventually_-small records the contrary rational-state obstruction for at least one positive shift. This is a conditional reduction only: no #251 irrationality, infinite rational-sum limit, universal producer, novelty, priority, significance, or external-review claim is made; #251 remains open. The actual-prime-gap specialization is distinct from coefficient_only_no_go and does not duplicate prime_gap_reformulation.

Result-atom population. 10 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. exact selected interface; 1 executable interface(s), 1 repository-registered selected result interface(s).

Publication placement. This family is also admitted to the short note.

- `ErdosProblems.Erdos251.primeGapTailShift_not_eventuallyIntegral_of_cofinal_small_mismatch`
- `ErdosProblems.Erdos251.not_eventuallyIntegralTailShift_of_cofinal_small_mismatch`
- `ErdosProblems.Erdos251.tailShift_not_both_integral_of_small_pair_of_digit_ne`
- `ErdosProblems.Erdos251.primeGap0_not_eventually_periodic`
- `ErdosProblems.Erdos251.rationalPrimeGapTail_has_positive_shift_not_eventually_small`

E.5 Coefficient only no go

Reader position. 51 of 70; display band: frontier. Formal editorial disposition: hold. These are separate classifications.

Reader entry. The exact finite carry identity and the two non-eventual-periodicity statements expose the coefficient-only barrier: irregularity and growth of a coefficient stream do not by themselves force irrationality, while the synthetic carry stream remains distinct from the actual prime-gap stream.

The exact finite carry identity and the two non-eventual-periodicity statements expose the coefficient-only barrier: irregularity and growth of a coefficient stream do not by themselves force irrationality, while the synthetic carry stream remains distinct from the actual prime-gap stream.

Authority and reach. Lean kernel plus authored synthesis; coefficient-only barrier; novelty and significance unassessed.

Exact boundary. CoefficientOnlyNoGo checks the exact finite partial-sum identity and the two non-eventual-periodicity statements. It does not assert the infinite limit of the synthetic countermodel, identify its coefficient stream with actual prime gaps, establish a prime-gap tail bridge, prove #251 irrationality, or make novelty, priority, significance, or external-review claims.

Result-atom population. 3 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. exact selected interface; 2 executable interface(s), 2 repository-registered selected result interface(s).

Publication placement. This family is also admitted to the short note.

- `ErdosProblems.Erdos251.carryPartialSum_natCast_eq`
- `ErdosProblems.Erdos251.carryCoeff_natCast_not_eventually_periodic`
- `ErdosProblems.Erdos251.primeGap0_not_eventually_periodic`

E.6 Totient shift propagation

Reader position. 60 of 70; display band: technical support. Formal editorial disposition: merge. These are separate classifications.

Reader entry. Odd rational denominators give totient-length integral shifts, and integrality propagates through the recurrence.

Odd rational denominators give totient-length integral shifts, and integrality propagates through the recurrence.

Authority and reach. Lean kernel; locally proved result; novelty unassessed.

Exact boundary. This describes rational states and supplies no contradiction for actual prime gaps.

Result-atom population. 2 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. represented by selected interface; 0 executable interface(s), 0 repository-registered selected result interface(s).

Publication placement. This family remains in the complete long record and is not a short-note headline.

- `ErdosProblems.Erdos251.tailShift_integral_totient_of_odd_den`
- `ErdosProblems.Erdos251.tailShift_integral_succ`
- `ErdosProblems.Erdos251.tailShift_integral_add`

References

[1] P. Erdős, *Sur certaines séries à valeur irrationnelle*, Enseign. Math. (2) **4** (1958), 93–100, doi:10.5169/seals-34629. The dyadic prime series is stated as unproved on p. 94; the factorial-prime family is stated on p. 93, with only the $k = 1$ proof printed on pp. 94–95; the variable-denominator classification, with nondecreasing denominators and a lower growth condition, is Theorem §3 on pp. 96–97.

[2] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monogr. Enseign. Math. 28, Geneva, 1980, p. 62.

[3] P. Erdős and C. Pomerance, *On the largest prime factors of n and $n + 1$*, Aequationes Math. **17** (1978), 311–321, doi:10.1007/BF01818569. The unnumbered dyadic irrationality theorem and its complete proof are in §7 on p. 320.

[4] P. Erdős, *On the irrationality of certain series: problems and results*, in A. Baker (ed.), *New Advances in Transcendence Theory*, Cambridge UP, 1988, pp. 102–109, doi:10.1017/CBO9780511897184.009.

[5] P. Erdős and E. G. Straus, *On the irrationality of certain Ahmes series*, J. Indian Math. Soc. (N.S.) **27** (1964), 129–133. MR 175848.

[6] K. Ford, B. Green, S. Konyagin, J. Maynard and T. Tao, *Long gaps between primes*, J. Amer. Math. Soc. **31** (2018), 65–105, doi:10.1090/jams/876. Theorem 1 on p. 66 gives the effective lower bound for the largest single consecutive-prime gap below X.

[7] A. Ya. Khinchin, *Continued Fractions*, University of Chicago Press, Chicago, 1964. Cited for the classical fact that a best approximation of the second kind is a convergent, which converts a certified partial-quotient prefix into a denominator lower bound.

[8] V. Kovač and T. Tao, *On several irrationality problems for Ahmes series*, Acta Math. Hungar. **175** (2025), 572–608.

[9] J. Land, *A conditional proof of the irrationality of $\sum_{n\geq 1} p_n 2^{-n}$ under a uniform Hardy-Littlewood prime-tuples conjecture*, research draft, 5 September 2026, with an accompanying Lean formalisation of the conditional argument. Its Conjecture 1 is the large-x form of Conjecture 1.3 of V. Kuperberg, *Sums of singular series with large sets and the tail of the distribution of primes*, arXiv:2210.09775v2, 15 June 2023.

[10] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in A. Platzer and G. Sutcliffe (eds.), CADE 28, Lecture Notes in Comput. Sci. 12699, Springer, 2021, pp. 625–635, doi:10.1007/978-3-030-79876-5_37.

[11] The mathlib Community, *The Lean mathematical library*, in CPP 2020, ACM, 2020, pp. 367–381, doi:10.1145/3372885.3373824. The article describes a December 2019 Lean 3-era snapshot; the current revision is the one named in the repository lock.

[12] J. Maynard, *Small gaps between primes*, Ann. of Math. (2) **181** (2015), 383–413.

[13] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I: Classical Theory*, Cambridge Stud. Adv. Math. 97, Cambridge UP, 2007, Chapter 6, pp. 168–198; Theorem 6.9, pp. 179–181, and Exercise 6.2.5, p. 183, doi:10.1017/CBO9780511618314.

[14] J.-C. Schlage-Puchta, *The irrationality of some number theoretical series*, arXiv:1105.1451, 2011. Theorem 2 treats base-b digit strings that concatenate the representations of a slowly growing integer sequence; Theorem 3 gives the $\mathbb{Q}$-linear independence of $1, S_0, S_1, \ldots$ with $S_k = \sum_{n\geq 1} p_n^k/n!$; Lemma 4, on pp. 5–6, states that for every nonzero $F \in \mathbb{Z}[x_0, \ldots, x_k]$ the equation $F(\delta_n, \ldots, \delta_{n+k}) = 0$ holds for a set of n of density zero, where $\delta_n = p_{n+1} - p_n$, and its proof uses Selberg's sieve.

[15] Y. Zhang, *Bounded gaps between primes*, Ann. of Math. (2) **179** (2014), 1121–1174.

[16] T. F. Bloom, *Erdős Problem #251*, `erdosproblems.com/251`, accessed 6 September 2026 (page displays "last edited 28 September 2025"). The current record labels the main dyadic problem open, cites `[Er58b]`, `[ErGr80, p. 62]` and `[Er88c, p. 103]`, attributes the whole factorial-prime family to the 1958 article, and explicitly describes its status as the website owner's present assessment, with no guarantee of literature completeness; it does not mention the 2026 counterexample in [18] to the adjacent variable-denominator conjecture.

[17] *Erdős Problem #251 discussion thread*, `erdosproblems.com/forum/thread/251`, accessed 6 September 2026. Cited for T. Tao's comment of 17:17 on 7 October 2025 recording the summation-by-parts equivalence and naming a uniform quantitative prime-tuples hypothesis as a plausible route, and for the two comments of J. Land of 6 September 2026 announcing [9]. Comments on that site are the responsibility of their authors and are not verified for correctness by the catalogue.

[18] ChatGPT 5.4 Pro (orchestrated by V. Kovač), *On the Erdős problem #251*, unpublished note, 2026, hosted by the Department of Mathematics, University of Zagreb, `web.math.pmf.unizg.hr`, accessed 6 September 2026; posted as a thread comment at 11:13 on 15 April 2026 [17].

[19] The Formal Conjectures Authors, *FormalConjectures.ErdosProblems.*`251`, Lean source at commit `f776d2f`, 2025, accessed 28 July 2026.